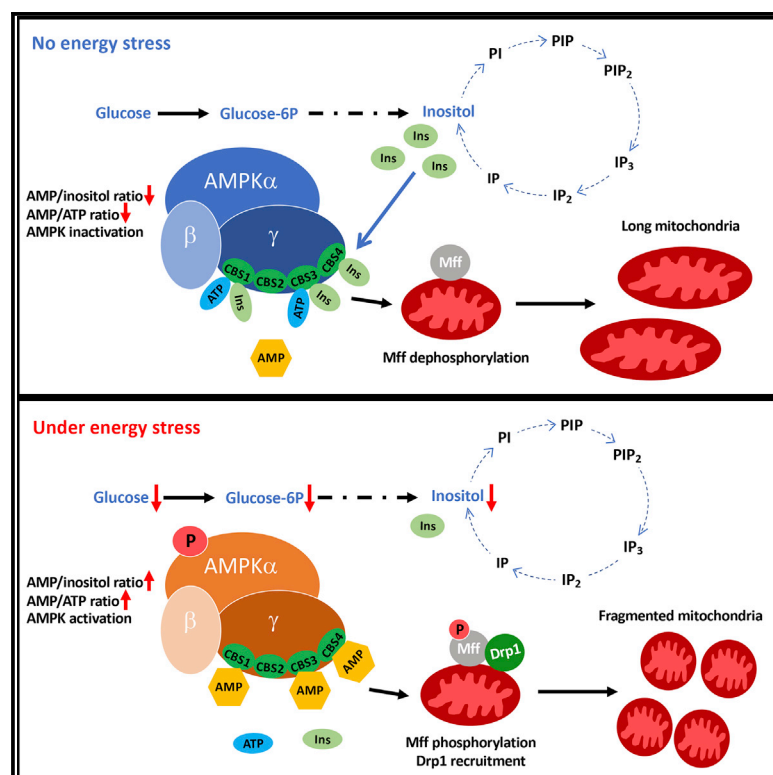


Inositol serves as a natural inhibitor of mitochondrial fission by directly targeting AMPK

Graphical abstract



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In brief

Hsu et al. demonstrate that inositol serves as an endogenous suppressor to limit AMPK activation and mitochondrial fission through its direct binding to the AMPK γ subunit. AMP/inositol and AMP/ATP ratios serve as critical determinants for AMPK activation and mitochondrial fission. AMPK is therefore an inositol sensor orchestrating mitochondrial fission by detecting the inositol gradient in cells.

Highlights

- IMPA-derived inositol modulates mitochondrial fission
- Inositol restricts aberrant AMPK activation and mitochondrial fission
- Inositol decline leads to AMPK-dependent mitochondrial fission upon energy stress
- Inositol binds directly to AMPK γ and competes with AMP for AMPK γ binding



Article

Inositol serves as a natural inhibitor of mitochondrial fission by directly targeting AMPK

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SUMMARY

Mitochondrial dynamics regulated by mitochondrial fusion and fission maintain mitochondrial functions, whose alterations underline various human diseases. Here, we show that inositol is a critical metabolite directly restricting AMPK-dependent mitochondrial fission independently of its classical mode as a precursor for phosphoinositide generation. Inositol decline by IMPA1/2 deficiency elicits AMPK activation and mitochondrial fission without affecting ATP level, whereas inositol accumulation prevents AMPK-dependent mitochondrial fission. Metabolic stress or mitochondrial damage causes inositol decline in cells and mice to elicit AMPK-dependent mitochondrial fission. Inositol directly binds to AMPK γ and competes with AMP for AMPK γ binding, leading to restriction of AMPK activation and mitochondrial fission. Our study suggests that the AMP/inositol ratio is a critical determinant for AMPK activation and establishes a model in which AMPK activation requires inositol decline to release AMPK γ for AMP binding. Hence, AMPK is an inositol sensor, whose inactivation by inositol serves as a mechanism to restrict mitochondrial fission.

INTRODUCTION

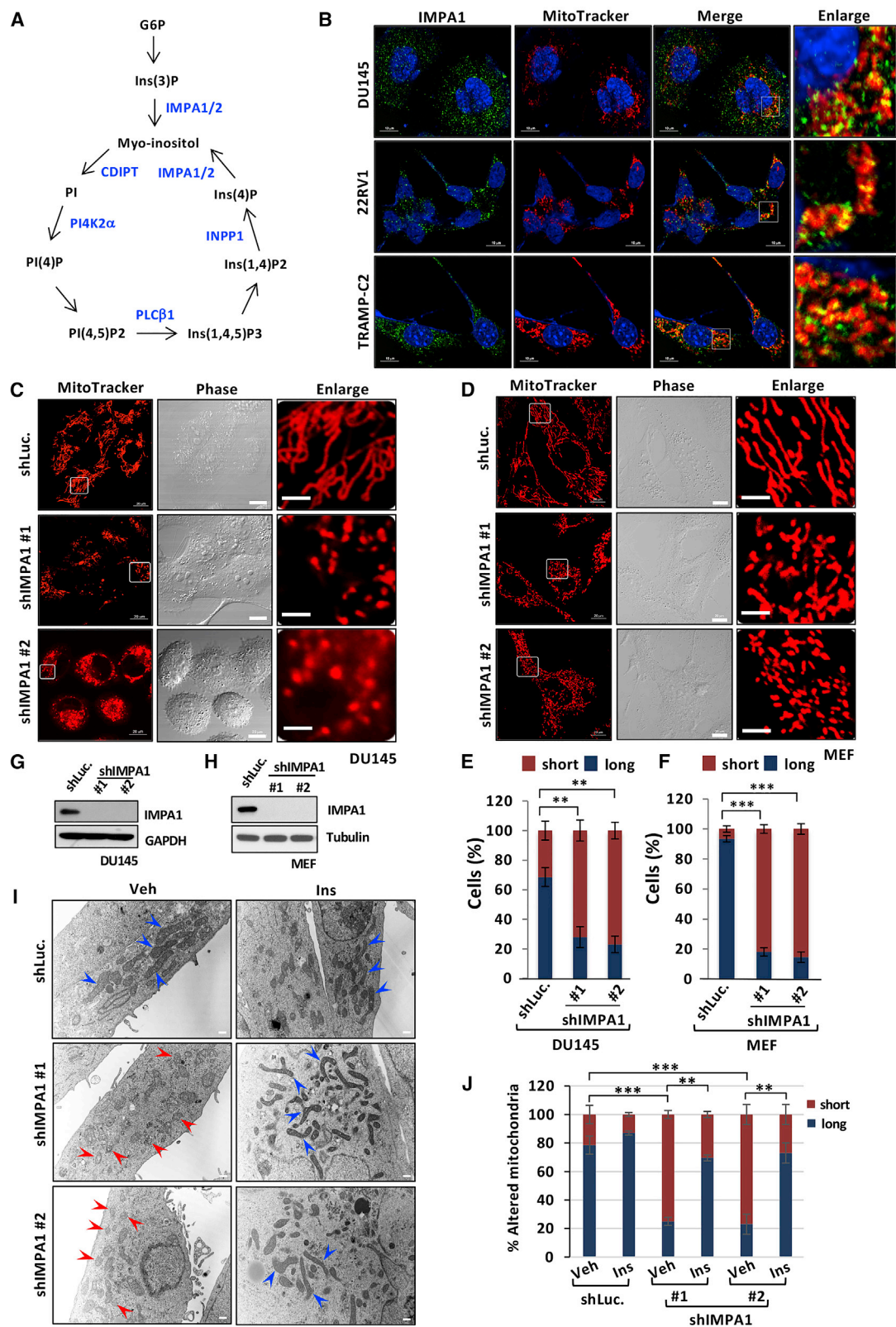
Phosphoinositides, phosphorylated forms of phosphatidylinositol (PI), play crucial roles in diverse signaling events and biological processes (Di Paolo and De Camilli, 2006). Phosphoinositides such as phosphatidylinositol 4-phosphate, PI(4)P, and phosphatidylinositol (4,5)-bisphosphate, PI(4, 5)P₂, are critical for controlling protein recruitment, membrane dynamics, and ion channel activity (Fair and Grinstein, 2012; Hammond et al., 2012). Phosphoinositide metabolism from PI(4, 5)P₂ generates important second messengers such as inositol (1, 4, 5) triphosphates (IP₃s) and diacylglycerol (DAG), which induce calcium release (Combettes and Champeil, 1994) and protein kinase C activation (Asaoka et al., 1992), respectively, to orchestrate apoptosis and cell growth. Myo-inositol (hereafter referred to as inositol) is a precursor used for the production of PI and phosphoinositides. The primary functions of inositol

are attributed to its conversion to PI and/or distinct phosphoinositides.

The defect of inositol metabolism is associated with pathological disorders including insulin resistance (Asplin et al., 1993), kidney failure (Holub, 1986), polycystic ovarian syndrome (PCOS) (Unfer et al., 2012), neurological disorders (Chalecka-Franaszek and Chuang, 1999; Jolles et al., 1993; Shetty et al., 1995), and reproductive problems (Carlomagno et al., 2011; Ciotta et al., 2011; Condorelli et al., 2011). In animal models of inositol-related knockout mice, numerous mitochondrion-related genes in the frontal cortex are upregulated similarly to mice treated with lithium that attenuates PI signaling in accordance with DNA microarray study (Toker et al., 2014), indicating that inositol metabolism may play an important role in modulating mitochondrial function.

Mitochondria play a central role in energy production, redox balance and metabolism control, signaling regulation, survival,





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and proliferation (Porporato et al., 2018; Vyas et al., 2016). Mitochondrial metabolism and dynamics maintaining mitochondrial functions (Mishra and Chan, 2016) are critically regulated by two distinct processes including fission and fusion, which orchestrate the overall length and connectivity of mitochondria. Abnormal dynamics of mitochondria underline various human diseases such as cancer, neurologic diseases, and cardiovascular diseases (Archer, 2013; Youle and van der Bliek, 2012). Defective energy metabolism, exercise, and nitrogen starvation have been known to trigger a change in mitochondrial dynamics by regulating diverse mitochondrial fission and fusion proteins that maintain a healthy mitochondrial population (Mishra and Chan, 2016). Although earlier studies have demonstrated how mitochondrial fission and fusion proteins mechanistically modulate mitochondrial dynamic processes upon different physiological conditions (Cribbs and Strack, 2007; Dickey and Strack, 2011; Kashatus et al., 2015; Toyama et al., 2016), the crosstalk between cellular metabolites and mitochondrial dynamics under energy stress are largely unknown.

Inositol is a metabolite generated by two steps of enzymatic reactions (Figure 1A). The first reaction involves the conversion of glucose-6-phosphate (G6P), derived primarily from the first step of glucose metabolism, to inositol monophosphate (IP) (Loewus et al., 1980; Majumder et al., 1997). The second reaction gives rise to inositol by dephosphorylating IP through inositol monophosphatase (IMPA), which consists of two distinct isoforms, namely, IMPA1 and IMPA2 (Figure 1A). Interestingly, our immunofluorescence (IF) assay revealed that IMPA1 is localized mainly in mitochondria (Figures 1B and S1A), indicative of the potential role of IMPA1/inositol in regulating mitochondrial functions.

Accumulating evidence reveals that energy metabolism could regulate cellular functions beyond its role in energy production and building blocks. This is achieved through the metabolites, such as α -ketoglutarate (α -KG) generated from mitochondria/TCA cycle, which can directly activate key enzymes involved in epigenetics and transcription for cell differentiation and stem cell maintenance (Carey et al., 2015; Lu and Thompson, 2012). Moreover, glycolysis-derived lactate represses RIG-I-like receptor (RLR) signaling by directly targeting mitochondrial antiviral-signaling (MAVS), which coordinates the crosstalk between energy metabolism and antiviral signaling (Zhang et al., 2019). These studies define the unique role of the metabolites in regulating cellular functions through direct protein targeting.

Inspired by the earlier findings revealing the critical role of certain metabolites in directly targeting protein activity and/or functions, we hypothesize that inositol can function as a second messenger to directly target certain proteins, thereby participating in unique signaling events to regulate biological outcomes beyond its known role in PI cycle metabolism. Our study not only offers the novel insight into how the specific metabolite crosstalks with 5'-AMP-activated protein kinase (AMPK)-dependent mitochondrial fission but also underscores the novel role of inositol in serving as a key metabolite directly restricting AMPK activation to maintain mitochondrial dynamics.

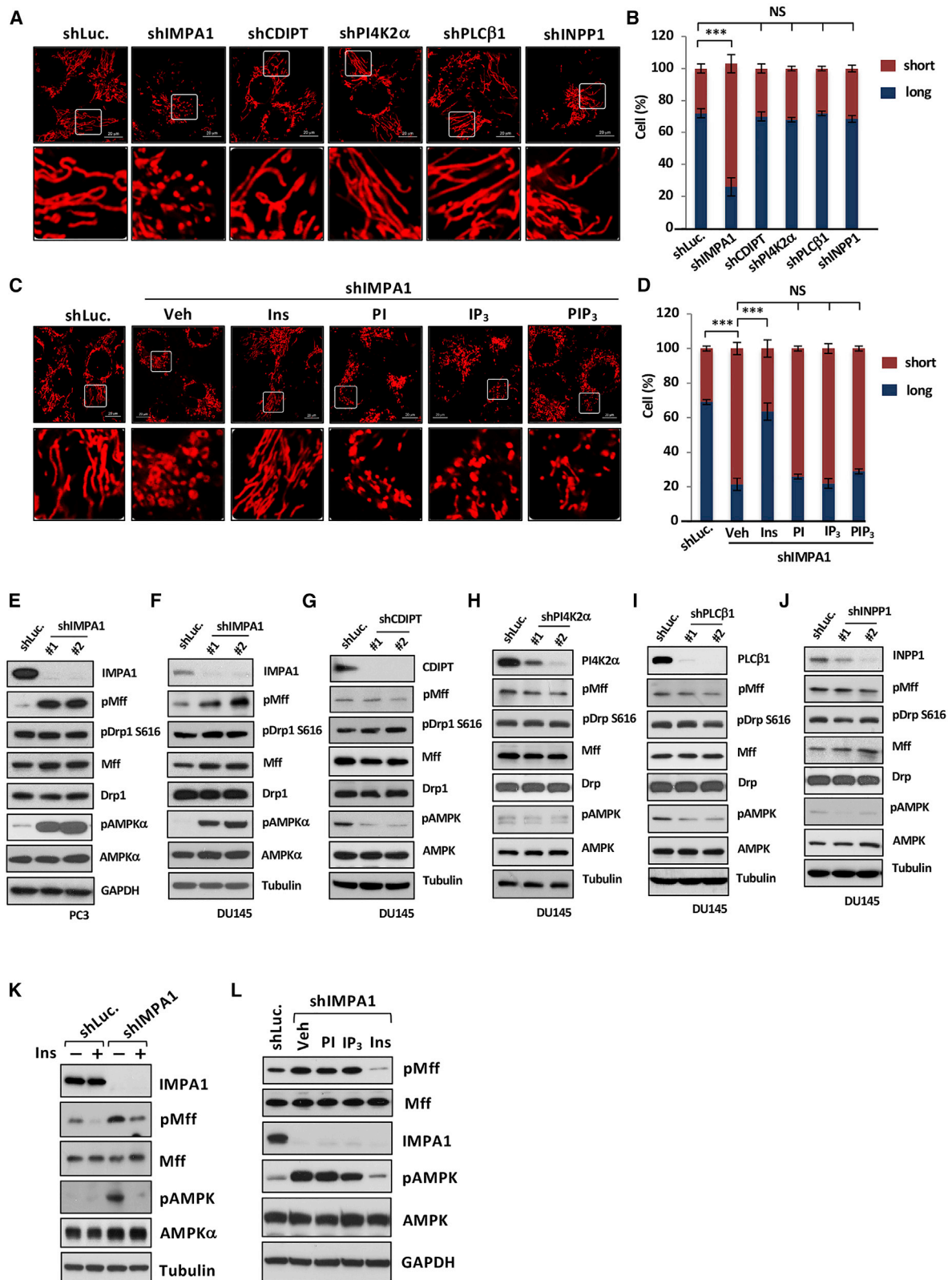
RESULTS

IMPA-derived inositol regulates mitochondrial fission

To explore a possible new role of inositol in cellular functions, we first focused on IMPA1, which is critical for inositol production in cells (Deranigh and Greenberg, 2009). An IF assay revealed that IMPA1 is localized mainly in mitochondria in diverse cancer cell lines (Figures 1B and S1A), which is indicative of the potential role of IMPA1/inositol in regulating mitochondrial functions. Because mitochondria are crucial for metabolic regulation, it is possible that specific metabolites generated from mitochondria may regulate mitochondrial dynamics. We then assessed whether IMPA1/inositol orchestrates mitochondrial dynamics. We examined the morphology of mitochondria with MitoTracker red staining by using a confocal microscope. Knockdown of IMPA1 in PC3 and DU145 cells and mouse embryonic fibroblasts (MEFs) markedly reduced inositol levels (Figures S1B and S1C); caused fragmented mitochondria; and decreased mitochondrial aspect ratio, interconnectivity, and elongation, indicative of mitochondrial fission (Figures 1C, 1E, 1G, and S1D–S1F). This phenomenon was also observed in MEFs with IMPA1 knockdown (Figures 1D, 1F, 1H, and S1G). Transmission electron microscopy (TEM) experiments confirmed that IMPA1 knockdown induced mitochondrial fission, reflected by shrunken and fragmented mitochondria (Figures 1I and 1J). IMPA1 knockout in DU145 cells by using CRISPR-Cas9 significantly caused mitochondrial fission, similar to IMPA1 knockdown (Figures S1H–S1K). Likewise, IMPA2 knockdown cells also exhibited a reduction in inositol level, fragmented mitochondria and reduced aspect ratio, interconnectivity, and elongation of mitochondria (Figures S1A–S1E).

Figure 1. IMPA1/inositol regulates mitochondrial fission

- (A) Phosphoinositide metabolism with a series of enzymes.
(B) Subcellular localization of IMPA1 using IMPA1 antibody (green), MitoTracker (red), and 4',6-diamidino-2-phenylindole (DAPI) (blue) in diverse cancer cell lines.
(C and D) Mitochondrial morphology by MitoTracker (red) in shLuc. or shIMPA1 DU145 cells (C) or MEFs (D). Scale bars, 20 μ m in MitoTracker and phase and 5 μ m in enlarge panel.
(E and F) Quantification of the mitochondrial morphology of the DU145 cells (C) and MEFs (D) shown in (E) and (F). Data are shown as the mean $\pm$ SEM of three independent experiments with 100 cells counted for each replicate. **, $P < 0.001$; ***, $P < 0.0001$ by two-tailed unpaired t-test.
(G and H) The shLuc. or shIMPA1 DU145 or MEFs were subjected to immunoblotting.
(I) TEM was performed in shLuc. or shIMPA1 DU145 cells of upon vehicle (Veh) or inositol treatment (Ins). Scale bars, 500 nm. Blue arrow, healthy/elongated mitochondria; red arrows, fragmented mitochondria.
(J) Quantification of the mitochondrial morphology (long and short) shown in (I). At least 50 mitochondria from DU145 cells of shLuc. or shIMPA1 (#1, #2) upon vehicle (Veh) or inositol treatment (Ins) and three independent experiments were analyzed in each group by the observer blinded to experimental conditions. **, $P < 0.001$, ***, $P < 0.0001$ by two-tailed unpaired t-test.



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It has been documented that W219 and D220 residues of IMPA are essential for IMPA phosphatase activity in bacteria (Goswami et al., 2018). We generated two mutants of W219A and D220A of human IMPA1 and performed a rescue experiment. Of note, restoration of IMPA1 and IMPA1 W219A, but not W220, rescued inositol levels in IMPA1 knockout cells (Figure S2F), suggesting that IMPA1 D220 is indispensable for phosphatase activity of IMPA1 in human cells. As a result, restoration of IMPA1 and IMPA1 W219A, but not IMPA1 D220A, reversed mitochondrial fission in IMPA1 knockout cells, indicative of the crucial role of IMPA1 phosphatase activity in repressing mitochondrial fission (Figures S2G–S2I). Similarly, pharmacological inactivation of IMPA1/2 activity by IMPA inhibitor lithium chloride (Cryns et al., 2008) also induced mitochondrial fission (Figures S3A and S3B). We ruled out the possibility that lithium chloride may act through GSK3 β inhibition to induce mitochondrial fission because the specific GSK3 β inhibitor BIO (Ding et al., 2019; Fang et al., 2019) that inhibited GSK3 β kinase activity similarly to lithium chloride failed to induce mitochondrial fission (Figures S3A–S3C).

Inositol directly regulates mitochondrial fission independently of its intermediates of the PI cycle

Metabolomics and biochemical assays revealed that the levels of inositol and its downstream intermediates, such as inositol (Ins), PI, PI 3-phosphate (PIP), PI 4,5-bisphosphate (PIP₂), PI 3,4,5-bisphosphate (PIP₃), inositol trisphosphate (IP₃), inositol biphosphate (IP₂), and IP were decreased in IMPA1 knockdown cells (Figures S1B and S1C). Intriguingly, knockdown of enzymes involved in PI cycles downstream of IMPA1/2 including CDP-DAG-inositol 3-phosphatidyltransferase (CDIPT), PI 4-kinase type 2 α (PI4K2 α), phospholipase C β 1 (PLC β 1), and inositol polyphosphate-1-phosphatase (INPP1), which catalyze the production of inositol downstream metabolites PI, PI(4)P, I(1,4,5)P₃, and I(4)P, did not induce mitochondrial fission (Figures 2A, 2B, and 2F–2J). Hence, inositol likely acts as a second messenger to directly restrict mitochondrial fission.

We found that the physiological levels of inositol are about 388 μ M in MEFs, 773 μ M in PC3 cells, 963 μ M in 22RV1 cells, and 963 μ M in TRAMP-C2 cells (Figure S2D). Moreover, the endogenous levels of inositol in kidney where the inositol is mainly produced is 653 μ M and 179 μ M in mouse serum (Figure S3D). Additionally, the physiological concentration for the metabolites such as PI, PI(3,4,5)P₃, and I(1,4,5)P₃ is below 10 μ M (Ando et al., 2003; Liang et al., 2014; McLaughlin et al., 2002). A mass-spectrometry-based analysis revealed that the treatment of inositol and its downstream metabolites in IMPA1 knockdown cells rescued intracellular inositol or phosphoinositide levels to the levels similar to control knockdown cells (Fig-

ures S3E and S3F). Of note, adding 25 μ M of inositol, but not 4 μ M of PI, I(1,4,5)P₃, and PI(3,4,5)P₃, in IMPA1 knockdown cells reversed the mitochondrial fission phenotype (Figures 2C, 2D, 1I, and 1J). Similarly, inositol also abrogated mitochondrial fission induced by lithium chloride (Figures S3G and S3H).

Mitochondrial fission commences upon the induction of the phosphorylation of dynamin-related protein 1 (Drp1) at S616 or mitochondrial fission factor (Mff) at S146 (Ducommun et al., 2015; Kashatus et al., 2015). In contrast, phosphorylation of Drp1 at S637 (pS637Drp1), which is induced and reduced by protein kinase A (PKA) (Chang and Blackstone, 2007), negatively regulates mitochondrial fission. We observed that the phosphorylation of Mff at S146 (pMff), but not phosphorylation of Drp1 at S616 (pS616Drp1), was markedly elevated in both IMPA1 and IMPA2 knockdown cells (Figures 2E, 2F, and S2E). Knockdown of CDIPT, PI4K2 α , PLC β 1, and INPP1 failed to induce either pMff or pS616Drp1 (Figures 2G–2J).

However, pS637Drp1 and CREB S133 phosphorylation, indicative of PKA activity, was not affected by IMPA1 knockdown (Figure S4A), suggesting IMPA1/inositol-mediated mitochondrial fission is unlikely to act through PKA and calcineurin signaling pathways. Moreover, proteolytic processing of OPA1, which functions as mitochondrial fusion (Head et al., 2009; Ishihara et al., 2006), was not significantly changed upon IMPA1 knockdown (Figure S4A). Remarkably, add-back of inositol, but not of its downstream metabolites, PI and IP₃, reversed the pMff (S146) levels in IMPA1 knockdown cells or IMPA1 knockdown cells permeabilized with streptolysin O (SLO), although an equal permeability to each metabolite was shown (Figures 2K–2L, S4B, and S4C). As a major receptor for Drp1, pMff (S146) recruits Drp1 to the mitochondrial outer membrane during mitochondrial fission (Losón et al., 2013; Otera et al., 2010). Consistently, Drp1 was recruited to and co-localized with mitochondria in IMPA1 knockdown cells (Figures S4D and S4E). Of note, Drp1 recruitment to mitochondria in IMPA1 knockdown cells was compromised by inositol restoration (Figures S4D and S4E). Collectively, IMPA-derived inositol serves as a repressor for pMff (S146) induction and mitochondrial fission independently of its conversion to downstream metabolites.

Aberrant mitochondrial fission leads to mitochondrial depolarization, autophagy/mitophagy, and/or apoptosis (Twig et al., 2008; Twig and Shirihai, 2011; Youle and Karbowski, 2005). Consistently, mitochondrial membrane potential was decreased in IMPA1 knockdown cells (Figures S5A and S5B). Notably, restoration of IMPA1, but not IMPA1 D220A, reversed mitochondrial depolarization in IMPA1 knockout cells (Figure S5C). Mitochondrial oxygen consumption (OCR) was also reduced in IMPA1 knockdown cells, and such a reduction was reversed by inositol restoration (Figure S5D). IMPA1 knockdown cells

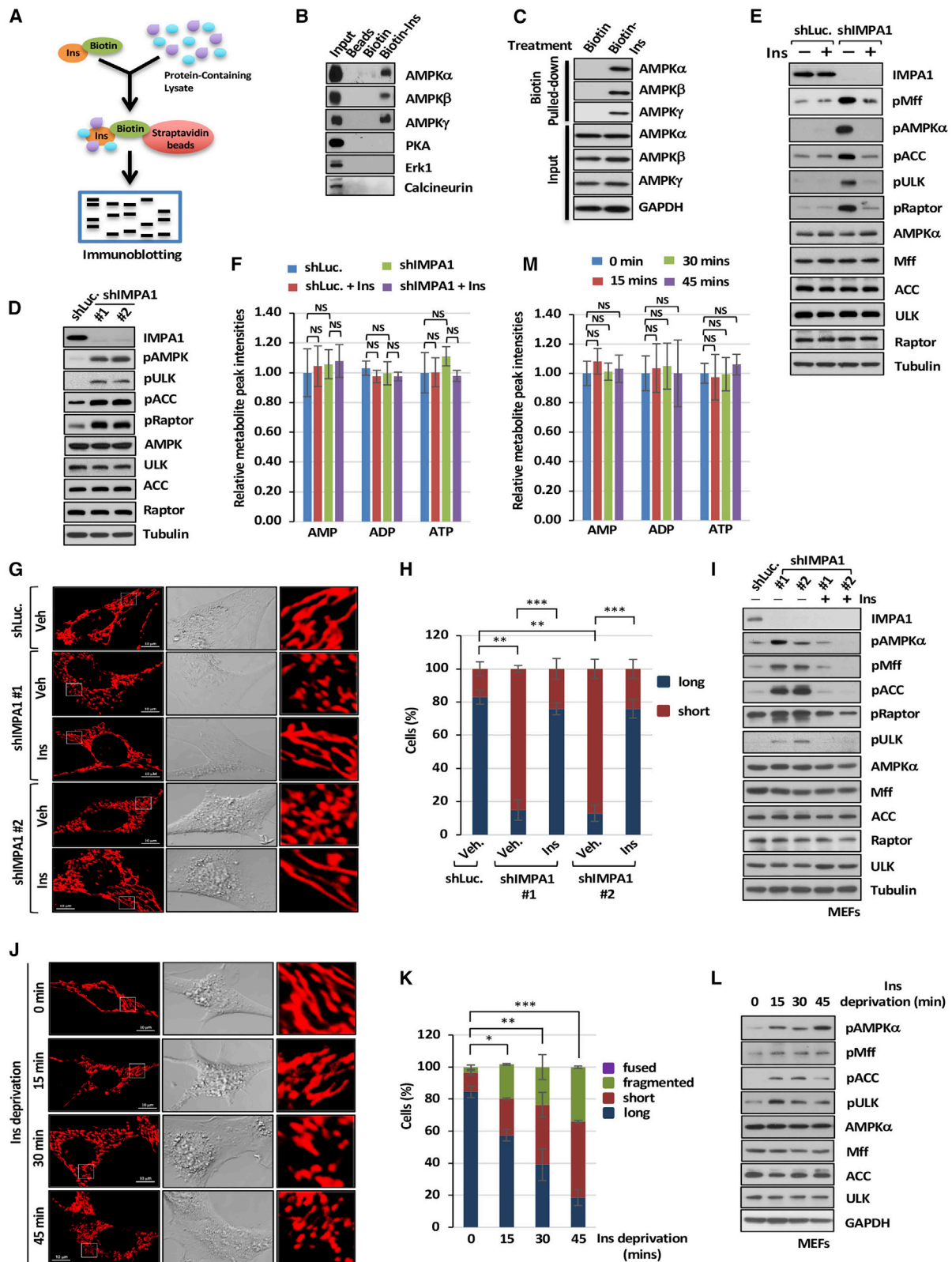
Figure 2. Inositol directly regulates mitochondrial fission independently of its intermediates of the phosphatidylinositol cycle

(A–D) Confocal images of mitochondrial morphology by MitoTracker (red) in diverse knockdown DU145 cells (A). Scale bars, 20 μ M. (C) Confocal images of mitochondrial morphology in shLuc. or shIMPA1 DU145 cells upon indicated metabolite treatment. Veh, vehicle. (B and D) Quantification of the mitochondrial morphology of the cells shown in (A) and (C). Data are shown as the mean $\pm$ SEM of three independent experiments with 100 cells counted for each replicate. ***, $P < 0.0001$; NS, non-significant by two-tailed unpaired t-test.

(E–J) Immunoblotting of PC3 or DU145 cells stably expressing shLuc. or specific short hairpin RNAs (shRNAs) for diverse genes.

(K) Immunoblotting of shLuc. or shIMPA1 DU145 cells with or without 25 μ M inositol treatment.

(L) Immunoblotting of shLuc. or shIMPA1 PC3 cells upon indicated metabolite treatment.



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also displayed autophagy judged by the increased LC3II level, which was antagonized by inositol treatment (Figure S5E). However, IMPA1 knockdown did not affect the basal level of reactive oxygen species and cell survival, although it induced mitochondrial fission (Figures S5F–S5I). Hence, inositol plays an important role in maintaining mitochondrial health and homeostasis.

Inositol binds to and inactivates AMPK activity, leading to restricting mitochondrial fission

The finding that inositol, but not its downstream metabolites, restricts mitochondrial fission indicates that inositol may have its direct protein targets, which have not yet been identified. We sought to identify direct targets of inositol responsible for mitochondrial fission regulated by inositol. To achieve this goal, we synthesized biotin-labeled inositol and mixed biotin or biotin-labeled inositol with PC3 cell extracts, followed by biotin-streptavidin pull-down (Figure 3A). We focused on the key kinase and phosphatase involved in mitochondrial fission including PKA (Dickey and Strack, 2011), calcineurin (Cribbs and Strack, 2007), Erk1 (Kashatus et al., 2015), and AMPK (Toyama et al., 2016). Of note, inositol selectively interacted with the AMPK complex including AMPK α , AMPK β , and AMPK γ , but not with other kinases or phosphatases involved in mitochondrial fission (Figure 3B). Moreover, inositol also failed to interact with other proteins involved in mitochondrial fission/fusion, such as Drp1, Fis1, Mff, MiD51, Opa1, Mfn1 and Mfn2, and autophagy/mitophagy, like LC3, LAMTOR2, and mTOR (Figure S6A).

To determine whether the interaction between inositol and AMPK occurs in cells *in vivo*, we treated cells with biotin or biotin-labeled inositol. A streptavidin pull-down assay revealed that inositol could interact with AMPK α , AMPK β , and AMPK γ in cells (Figure 3C). Using the immunoprecipitation assay with AMPK α antibody under glucose-proficient or glucose-deprived conditions or with a (hemagglutinin) HA antibody, we detected high levels of endogenous inositol in the AMPK α immunocomplex under glucose-containing conditions and in HA-AMPK α overexpression immunocomplex (Figures S6B and S6C), indicative of the *in vivo* interaction between inositol and AMPK α . Moreover, lower levels of endogenous inositol were detected in AMPK α immunocomplex upon glucose deprivation (–Glc.) compared with glucose-proficient conditions (+Glc.) (Figure S6B), suggesting that AMPK activation upon glucose

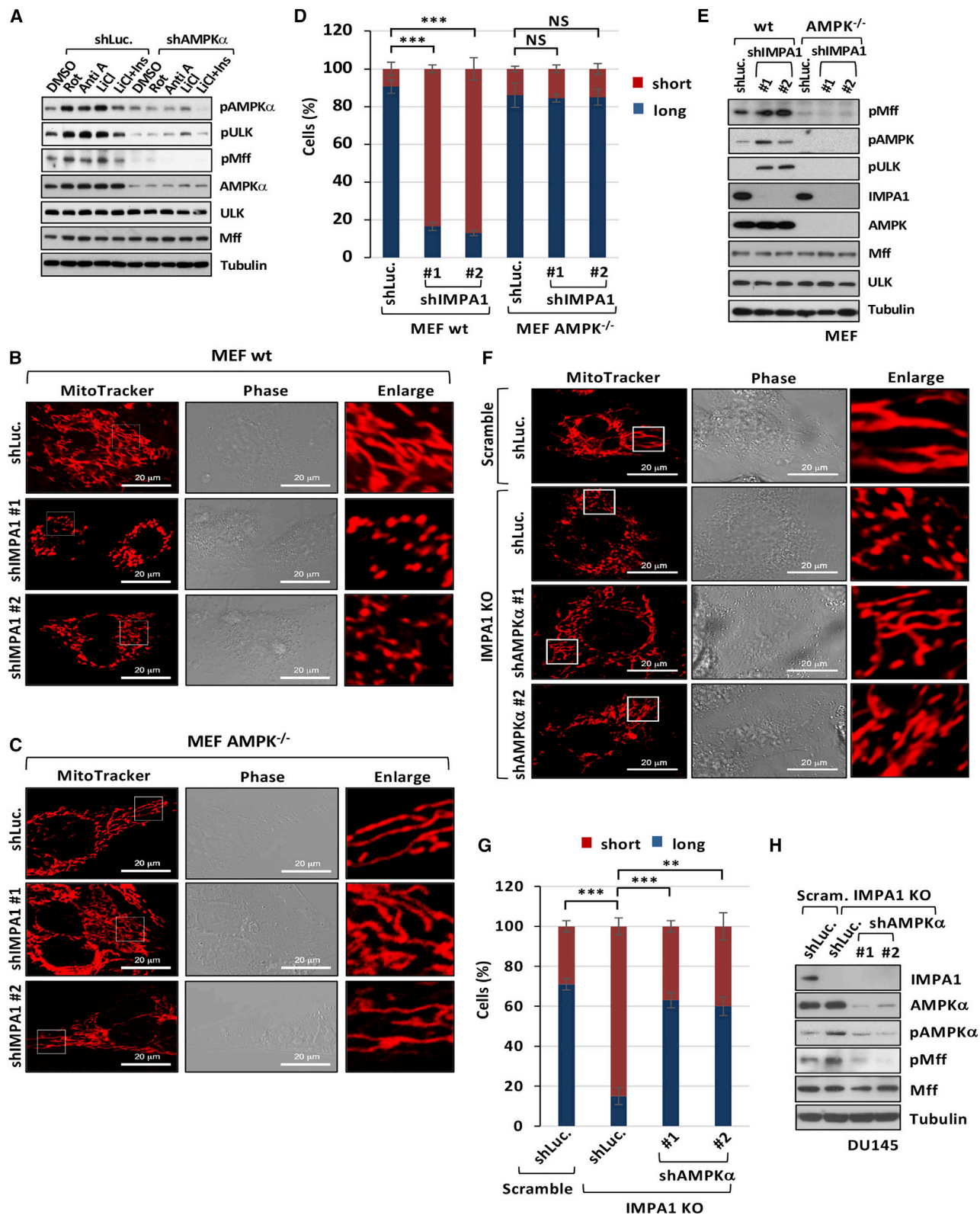
deprivation is correlated with lower inositol binding to AMPK than AMPK inactivation under glucose-proficient conditions.

AMPK activation induced by energy stress inducers such as mitochondria-damaging agents elicits mitochondrial fission through directly phosphorylating Mff at S146 (pMff) (Ducommun et al., 2015; Toyama et al., 2016). Increased AMP level following the decline in ATP level is thought to be a critical event for AMPK activation under distinct stress conditions (Hsu et al., 2021; Steinberg and Carling, 2019). Our finding that inositol decline upon the loss of IMPA1 or IMPA2 induces pMff (S146) and mitochondrial fission and that inositol interacts with AMPK suggests that inositol is likely a direct suppressor of AMPK. Indeed, both IMPA1 and IMPA2 knockdown cells displayed markedly enhanced AMPK activation, reflected by elevated phosphorylation of AMPK at T172 and phosphorylation of its downstream substrates (Figures 3D, 2E, 2F, S1K, and S6D). Inactivation IMPA1/2 by lithium chloride, but not GSK3 β inhibitor BIO, also elevated phosphorylation of AMPK and its downstream substrate Mff (Figure S6E). Moreover, restoration of IMPA1, but not IMPA1 D220A, reversed AMPK and Mff phosphorylation in IMPA1 knockout cells (Figure S2I), further supporting that phosphatase activity of IMPA1 is crucial for suppressing mitochondrial fission. Restoration of inositol, but not PI and IP₃, in IMPA1 knockdown, IMPA1 knockdown SLO-permeabilized, or lithium-chloride-treated cells attenuated phosphorylation of AMPK and its substrate Mff (Figures 3E, 2K, 2L, and S6E), which is similar to the treatment of AMPK inhibitor compound C (Figure S6F). However, knockdown of CDIPT, PI4K2 α , PLC β 1, and INPP1 failed to induce phosphorylation of AMPK and Mff (Figures 2G–2J). Surprisingly, AMP, ADP, and ATP levels; AMP/ATP ratio or ADP/ATP ratio in IMPA1 knockdown cells; or inositol-restored IMPA1 knockdown cells remained unchanged compared with those of control knockdown cells (Figures 3F and S6G–S6L).

Because our above findings were conducted in an inositol-containing medium, we asked the question whether the loss of IMPA1 also triggers AMPK activation and mitochondrial fission in inositol-free medium. We indeed reproduced our findings in an inositol-free medium that inositol decline upon IMPA1 knockdown triggers mitochondrial fission and AMPK activation (Figures 3G–3I). Remarkably, restoration of inositol completely impaired aberrant mitochondrial fission, AMPK activation, and

Figure 3. Inositol binds to AMPK and regulates AMPK activity and mitochondrial fission

- (A) Biotin-labeled inositol (Biotin-Ins) *in vitro* binding assay was performed by immunoblotting analysis.
(B) Immunoblotting of biotin pull-down assay in PC3 cells with indicated antibodies.
(C) Immunoblotting of biotin pull-down assay upon 25 μ M of biotin or Biotin-Ins treatment for 1 h.
(D) Immunoblotting of shLuc. or shIMPA1 PC3 cells.
(E) Immunoblotting with shLuc. or shIMPA1 PC3 cells upon vehicle or Ins treatment.
(F) Metabolic profiling of AMP, ADP, and ATP in DU145 cells expressing shLuc. or shIMPA1 upon vehicle (Veh) or Ins treatment was performed by targeted mass spectrometry analysis. NS, non-significant by two-tailed unpaired t-test.
(G) Confocal images of mitochondrial morphology in shLuc. or shIMPA1 MEFs upon Veh. or Ins treatment. Scale bars, 10 μ m.
(H) Quantification of the mitochondrial morphology of the MEFs shown in (G). Data are shown as the mean $\pm$ SEM of three independent experiments with 100 cells counted for each replicate. *, $P < 0.005$, **, $P < 0.001$; ***, $P < 0.0001$ by two-tailed unpaired t-test.
(I) Immunoblotting of shLuc. or shIMPA1 MEFs upon Veh. or Ins treatment.
(J) Confocal images of mitochondrial morphology in MEFs upon Ins deprivation. Scale bars, 10 μ m.
(K) Quantification of the mitochondrial morphology of the cells shown in (J) upon Ins deprivation. *, $P < 0.005$, **, $P < 0.001$ by two-tailed unpaired t-test.
(L) Immunoblotting of MEFs upon Ins deprivation.
(M) Metabolic profiling of AMP, ADP, and ATP in MEFs upon inositol deprivation was performed. NS, non-significant by two-tailed unpaired t-test.



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phosphorylation of its downstream substrates in inositol-free medium. Of note, inositol deprivation in MEFs or IMPA1 knockdown MEFs for 15 to 45 min triggered phosphorylation of AMPK and its substrates and mitochondrial fission without affecting ATP levels, suggesting that inositol decline is sufficient to induce AMPK activation and mitochondrial fission independently of ATP level (Figures 3J–3M and S6M). Hence, IMPA-derived inositol serves as an endogenous suppressor of AMPK activation in cells.

Consistent with the previous study (Mihaylova and Shaw, 2011), AMPK deficiency impaired the induction of mitochondrial fission as well as pMff and pULK levels upon the treatment of mitochondrial electron transport chain (ETC) inhibitors rotenone and antimycin A (Figure 4A). AMPK activation, Mff phosphorylation, and mitochondrial fission upon IMPA1 knockdown, IMPA1 knockout, or IMPA inactivation through lithium chloride were all compromised upon AMPK knockdown or knockout, accompanied by reduced pMff or pULK levels (Figures 4B–4E; Figures 4F–4H; Figures S3G and S3H). Decreased mitochondrial membrane potential induced by the loss of IMPA1/inositol was also rescued upon AMPK knockdown, AMPK inhibition by compound C, or AMPK knockout (Figures S5J–S5S). Thus, IMPA/inositol restricts mitochondrial fission and maintains mitochondrial membrane potential through inhibiting AMPK activation.

Inositol decline leads to AMPK activation and mitochondrial fission under diverse stresses

Glucose through glycolysis serves as a main source for the generation of G6P, which is a key precursor for inositol biosynthesis (Eisenberg, 1967). We rationalized that glucose would be a key nutrient for maintaining inositol levels in cells. Indeed, glucose deprivation or treatment with mitochondrion-damaging agents like rotenone or antimycin A reduced inositol levels, similar to results of IMPA1 knockdown or lithium chloride treatment (Figures S6N; Figures S1B and S1C), and correlated with reduced G6P levels (Figure S6O). As a result, glucose deprivation not only decreased inositol levels but also markedly reduced inositol-AMPK binding (Figure S6B). Because inositol serves as a barrier to restrict AMPK-dependent mitochondrial fission, we determined whether glucose represents an important physiological cue to restrict mitochondrial fission. By performing time-lapse microscopic experiments, we observed the occurrence of fragmented mitochondria indicative of mitochondrial fission 10 min after glucose deprivation, similar to rotenone treatment (Figures

5A–5D; Videos S1 and S3). This finding was correlated with AMPK activation (Figures 5E and 5F; Figures S6P and S6Q). Of note, inositol restoration reversed mitochondrial fragmentation induced by not only glucose deprivation but also rotenone or antimycin A treatment, which also caused the reduction of inositol levels in cells (Figures 5A–5D; Figure S6N; Videos S2 and S4), and correlated with reduced AMPK activation (Figures 5E and 5F; Figures S6P and S6Q). Thus, glucose is a key physiological cue to limit AMPK activation and mitochondrial fission by maintaining inositol levels.

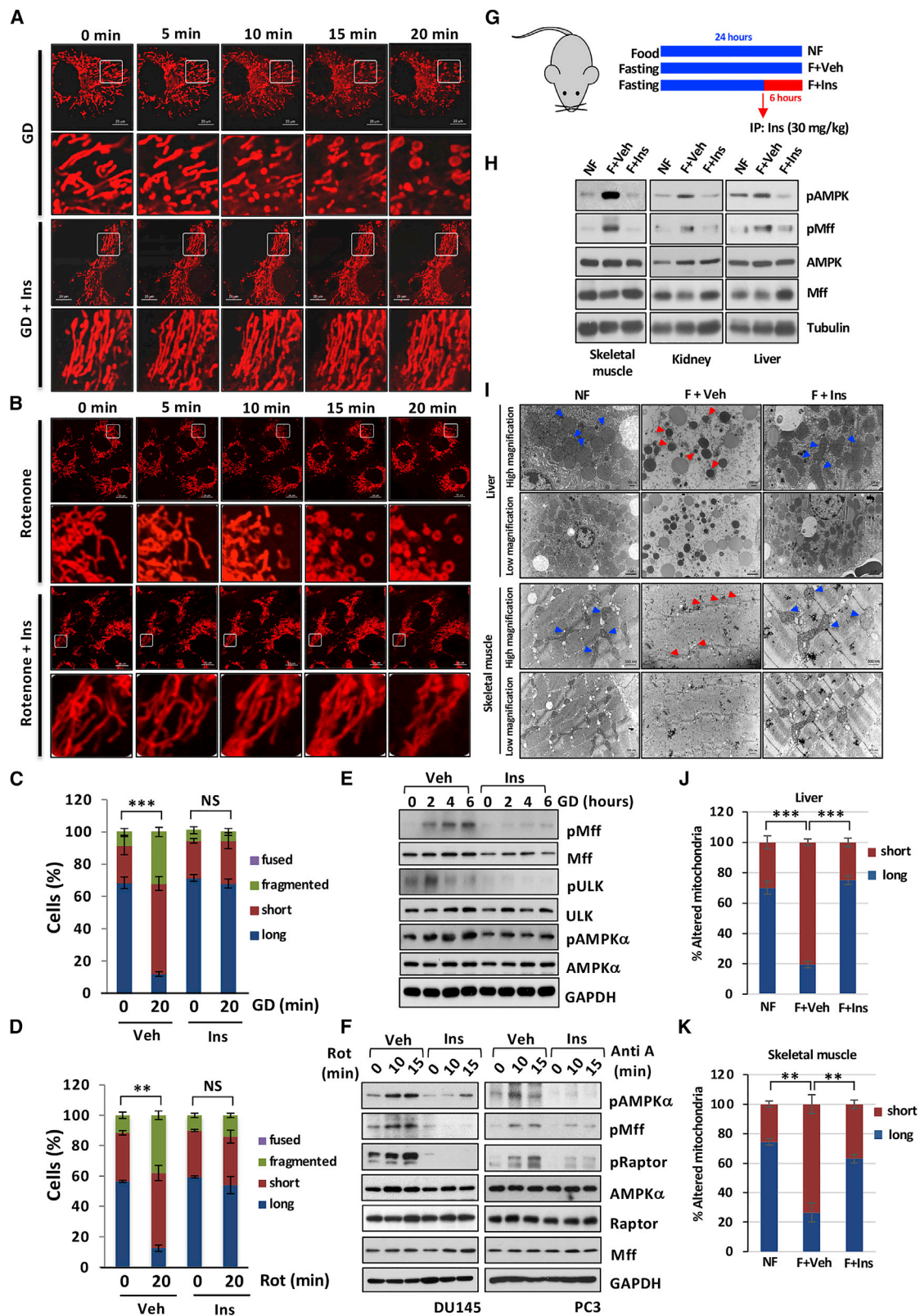
To further investigate the physiological role of inositol in repressing AMPK-dependent mitochondrial fission *in vivo*, we conducted an animal study in which mice were fasted for 24 h to reduce inositol levels (Figures S7A–S7C; Figures S7J and S7K), followed by intraperitoneal injection or oral administration of vehicle or inositol (Figures 5G and S7D). Inositol levels in tibialis anterior, liver, and kidney were partially rescued in fasted mice by either intraperitoneal injection or oral administration (Figures S7A and S7B; Figures S7J and S7K). TEM revealed that fragmented mitochondria were significantly increased in liver and skeletal muscle of fasted mice, which correlated with elevated phosphorylation of AMPK at T172 and phosphorylation of Mff at S146 (Figures 5H–5K; Figures S7E–S7I). Of note, inositol restoration in fasted mice abrogated the elevation of AMPK and Mff phosphorylation and mitochondrial fragmentation in skeletal muscle, kidney, and liver (Figures 5H–5K; Figures S7E–S7I). Hence, inositol is an endogenous inhibitor of AMPK-dependent mitochondrial fission.

CDIPT loss leading to inositol accumulation prevents mitochondrial fission upon energy stress

If inositol reduction in cells elicits spontaneous AMPK activation and mitochondrial fission, inositol accumulation in cells should exhibit an opposite effect on AMPK activation and mitochondrial fission under energy stress conditions. We tested this hypothesis by establishing cells with inositol accumulation. Because CDIPT is the enzyme converting inositol to PI, a loss of CDIPT may lead to inositol accumulation. Indeed, the loss of CDIPT caused inositol accumulation and abrogated AMPK activation, pMff induction, and mitochondrial fission upon glucose deprivation or antimycin A treatment (Figures 6A–6E). Of note, treatment of IMPA1/2 inhibitor lithium chloride could partially rescue the defects in AMPK activation, pMff induction, and mitochondrial fission in CDIPT knockdown cells upon glucose deprivation (Figures 6F and 6G; Figure S6R). AMPK activation upon the

Figure 4. Inositol decline upon IMPA1 loss induces mitochondrial fission in an AMPK-dependent manner

- (A) Immunoblotting of shLuc. or shAMPK α DU145 cells treated with DMSO, 250 ng/ml of rotenone (Rot), 10 μ M of antimycin A (Anti A), 500 μ M of LiCl, or 500 μ M of LiCl plus 25 μ M of Ins.
- (B and C) Confocal images of mitochondrial morphology in wild-type (WT) MEFs and AMPK $^{-/-}$ MEFs stably transduced with shLuc. or shIMPA1. Scale bars, 20 μ m.
- (D) Quantification of the mitochondrial morphology of MEFs shown in (B) and (C). Data are shown as the mean $\pm$ SEM of three independent experiments with 100 cells counted for each replicate. **, $P < 0.001$; ***, $P < 0.0001$; NS, non-significant by two-tailed unpaired *t*-test.
- (E) Immunoblotting of WT and AMPK $^{-/-}$ MEFs stably expressing shLuc. or shIMPA1.
- (F) Confocal images in CRISPR-Cas9 knockout DU145 cells stably transduced with the guide RNA pairs of scramble (Scram.), IMPA1 knockout clone #1 (IMPA1 KO), shLuc., or shAMPK α . Scale bars, 20 μ m.
- (G) Quantification of the mitochondrial morphology of the DU145 shown in (F). **, $P < 0.001$; ***, $P < 0.0001$ by two-tailed unpaired *t*-test.
- (H) Immunoblotting of CRISPR-Cas9 knockout DU145 cells transduced with the guide RNA pairs of Scram., IMPA1 KO, shLuc., or shAMPK α .



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treatment of AMPK activators, such as AICAR or A769662, induced pMff and mitochondrial fission, whereas inositol treatment inhibited AMPK activation, pMff induction, and mitochondrial fission induced by AICAR or A769662 (Figures S8A–S8E; Videos S5 and S6). Collectively, inositol decline triggers AMPK activation and mitochondrial fission under energy stress conditions.

AMPK activation plays a role in maintaining cell survival under metabolic stress (Jones et al., 2005). We speculated that IMPA1 knockdown cells with elevated AMPK activation may exhibit less cell death under metabolic stress than control knockdown cells. Indeed, the impairment of apoptosis, accompanied by a marked reduction in cleaved PARP, caspase 3, and elevated AMPK activation, upon glucose deprivation was observed in IMPA1 knockdown cells (Figures S9A–S9C), indicating that IMPA1/inositol regulates AMPK-mediated mitochondrial fission and cell survival under metabolic stress.

Inositol competes with AMP in binding to the AMPK γ subunit and represses AMPK kinase activity

AMP serves as a physiological activator of the AMPK complex by directly binding to the AMPK γ subunit, whereas ATP acts as a suppressor of AMPK by competing with AMP for binding to AMPK γ (Gowans et al., 2013). Hence, an increase of the AMP/ATP ratio results in AMPK activation during energy stress (Gowans et al., 2013). As inositol interacts with AMPK and serves as a key metabolite to restrict AMPK activation and mitochondrial fission in cells and in mice, we assessed whether inositol could directly inhibit AMPK kinase activity. The *in vitro* kinase assay showed that inositol inhibits kinase AMPK activity toward phosphorylation of two AMPK substrates, namely, GST-Mff or ULK protein (Figures 7A–7C; Figure S10A). We also performed an AMPK enzymatic kinetic curve analysis under different concentrations of SAMS peptide and inositol. K_i (519.8 μ M) for inositol on AMPK activity was obtained by fitting the Michaelis-Menten equation (Figures 7D and 7E), and inositol displayed non-competitive inhibition according to a Lineweaver-Burk plot (Figures 7D and S10B), suggesting that inositol potentially serves as an allosteric inhibitor of AMPK. Given that AMPK has multiple isoforms, we also examined the effects of inositol on the activity of different AMPK isoform complexes. The results revealed that inositol inhibits the activity of AMPK α 1 β 1 γ 1, AMPK α 1 β 2 γ 1, AMPK α 2 β 2 γ 1, and AMPK α 1 β 1 γ 3 complexes, but not the activity of AMPK α 2 β 1 γ 1 and AMPK α 1 β 1 γ 2 complexes (Figures 7B and 7C; Figures S10C–S10F). Remarkably, inositol,

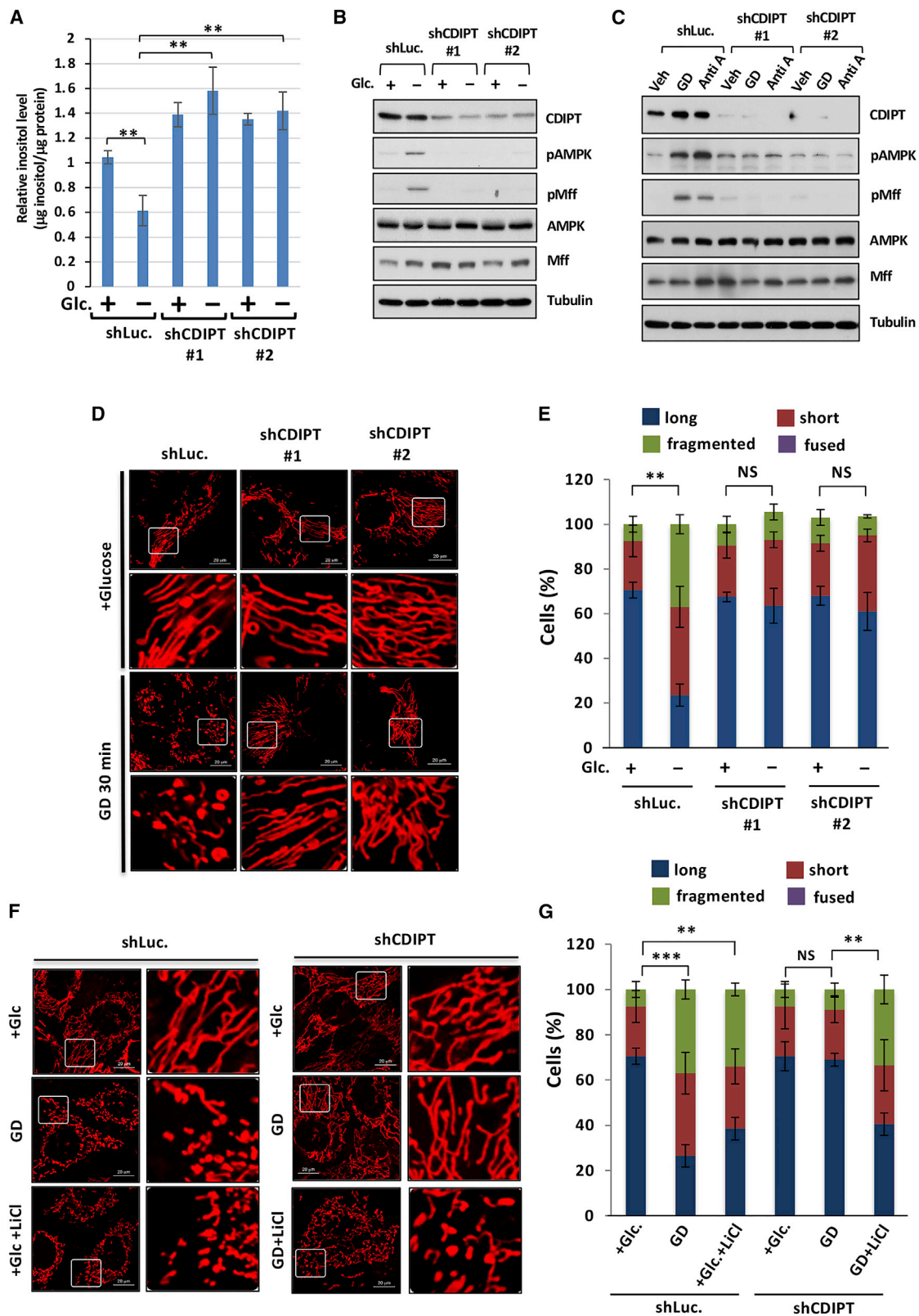
but not PI, IP $_3$, and IP, impaired AMPK kinase activity (Figures S10G and S10H). Moreover, AMP directly enhances AMPK kinase activity in a dose-dependent manner (Figures 7G and S10I).

To further elucidate whether both inositol and ATP are essential for inhibition of AMPK activity, we examined *in vitro* AMPK activity upon different concentrations of inositol and ATP. Although 50 and 500 μ M of inositol were sufficient to inhibit AMPK kinase activity, ATP displayed no significant inhibitory effect at these two concentrations on AMPK kinase activity (Figure 7E; Figures S10J and S10K). However, ATP at a physiological concentration (5 mM) achieved the inhibitory effect on AMPK kinase activity (Figure 7E). Under physiological concentrations of both inositol and ATP, inositol could cooperate with ATP to inhibit AMPK kinase activity (Figure 7F). Importantly, the inhibitory effect on AMPK kinase activity by inositol could be rescued by adding AMP, but not ATP (Figure 7G). Inositol also attenuated AMPK activity and AMPK kinase activation induced by a low concentration of AMP, whereas a high concentration (1 mM or above) of AMP impaired the inhibitory effect of inositol (0.5 mM) on AMPK kinase activity (Figures 7H and S10L), suggesting that inositol and ATP represent endogenous inhibitors of AMPK but inositol requires a much lower concentration than ATP to achieve its inhibitory effect on AMPK activation. Collectively, inositol not only serves as an allosteric inhibitor of AMPK by non-competitive inhibition with the substrate SAMS peptide but also represses AMP-mediated AMPK activation by competing with AMP for binding to AMPK γ .

An *in vitro* binding assay using recombinant proteins revealed that inositol could directly bind to AMPK γ 1, but not AMPK α 1 and AMPK β 1 (Figure 7I). A saturation binding assay using tritium-labeled inositol revealed that inositol directly binds to AMPK γ 1 in a dose-dependent manner (Figure 7J). Direct binding of inositol to AMPK γ 1 was also demonstrated by isothermal titration calorimetry (ITC) (Figures S10M–S10O). Inositol binds to Bateman domain 2 composed of cystathionine- β synthase 3 (CBS3) and CBS4 motifs, but not Bateman domain 1 composed of CBS1 and CBS2 motifs. Moreover, inositol could bind to single CBS1, CBS3, and CBS4 motifs of AMPK γ 1, but not CBS2 motif within its physiological concentration in cells (Figures 7K–7M). However, inositol bound to the CBS4 motif of AMPK γ 1 with a much higher affinity than CBS1 and CBS3 motifs of AMPK γ 1, as it at 40 μ M could bind to the CBS4 motif of AMPK γ 1, but not CBS1 and CBS3 motifs of AMPK γ 1 (Figures 7L and 7M).

Figure 5. Inositol decline leads to AMPK activation and mitochondrial fission under diverse stresses

- Time-lapse images of DU145 cells stained with MitoTracker upon glucose deprivation (GD) or GD with Ins treatment for various time points. Scale bars, 20 μ m.
- Time-lapse images of DU145 cells stained with MitoTracker upon indicated treatments. Scale bars, 20 μ m.
- Quantification of the mitochondrial morphology of the cells shown in (A) at 20 min upon indicated treatment. ***, $P < 0.0001$; NS, non-significant by two-tailed unpaired t -test.
- Quantification of the mitochondrial morphology of the cells shown in (B) at 20 min upon Rot with or without Ins treatment. **, $P < 0.001$; NS, non-significant by two-tailed unpaired t -test.
- Immunoblotting of DU145 cells upon indicated treatments.
- Immunoblotting of PC3 or DU145 cells upon indicated treatments.
- The graphic shows mice fasted for 24 h and Ins treatment by intraperitoneal injection (30 mg/kg) at mice fasted for 18 h.
- The protein extract from diverse tissues were subjected to immunoblotting.
- Images of TEM upon mice fed with food, no fasting (NF), fasted with vehicle (F+Veh), and fasted with Ins (F+Ins). Scale bars, 500 nm and 2 μ m.
- Quantification of the mitochondrial morphology (long and short) shown in (I). **, $P < 0.001$ by two-tailed unpaired t -test.



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The current model proposed that the binding of AMP to the AMPK γ subunit causes the conformational change of the AMPK complex that exposes AMPK α for the phosphorylation of AMPK α at T172 by its upstream kinases, leading to AMPK activation (Gowans et al., 2013). Thus, AMP serves as an allosteric activator for AMPK activation. Given that AMP, but not ATP, rescued impaired of AMPK kinase activity by inositol upon the addition of the same doses (50 and 125 μ M) *in vitro* (Figure 7G), we examined whether AMP abrogates the binding of inositol to AMPK γ 1. The result revealed that an increased amount of AMP, but not ATP, disrupted inositol-AMPK γ binding (Figure 7N). The competition binding assay showed that AMP, but not ATP, inhibits inositol binding to AMPK γ 1 in a dose-dependent manner (Figures 7O and 7P). Reciprocally, we observed that AMP bound to AMPK γ 1, but not AMPK α 1 and AMPK β 1, and such binding was markedly abolished by inositol (Figure 7Q). We conclude that inositol is a natural inhibitor of AMPK γ by directly binding to AMPK γ and preventing AMP from binding to AMPK γ .

DISCUSSION

Mitochondrial fission and fusion are tightly regulated to maintain normal mitochondrial functions and cellular homeostasis. Defects in these processes are linked to diverse human diseases (Archer, 2013; Youle and van der Bliek, 2012). Although numerous proteins have been identified in regulation of mitochondrial fission or fusion (Cárdenas et al., 2010; Cogliati et al., 2013; Losón et al., 2013; Richter et al., 2014), the specific metabolites impinging on mitochondrial dynamics are largely unknown. Our study identifies inositol, generated from IMPA1 and/or IMPA2 localized in mitochondria, as a key metabolite that orchestrates mitochondrial dynamics. We show that a decline in inositol levels during metabolic stress drives mitochondrial fission in cells and mice, leading to muscle atrophy (Figure 5; Figure S7). Hence, the fact that an imbalance of inositol level leads to defects in mitochondrial dynamics provides great insight into the pathogenesis of mitochondrial fission and fusion in human disease (Archer, 2013).

Earlier studies suggested that the functional role of inositol is primarily attributed to the generation of its downstream metabolites. However, the direct role of inositol in serving as a second messenger to regulate signaling and biological outcomes has not been reported. We present convincing evidence that inositol serves as an allosteric inhibitor that represses AMPK activation to restrict mitochondrial fission independently of its known role in serving as a precursor for PI and phosphoinositides. In addition, inositol also regulates AMPK activation to regulate cell death features under metabolic stress (Figures S9A–S9C),

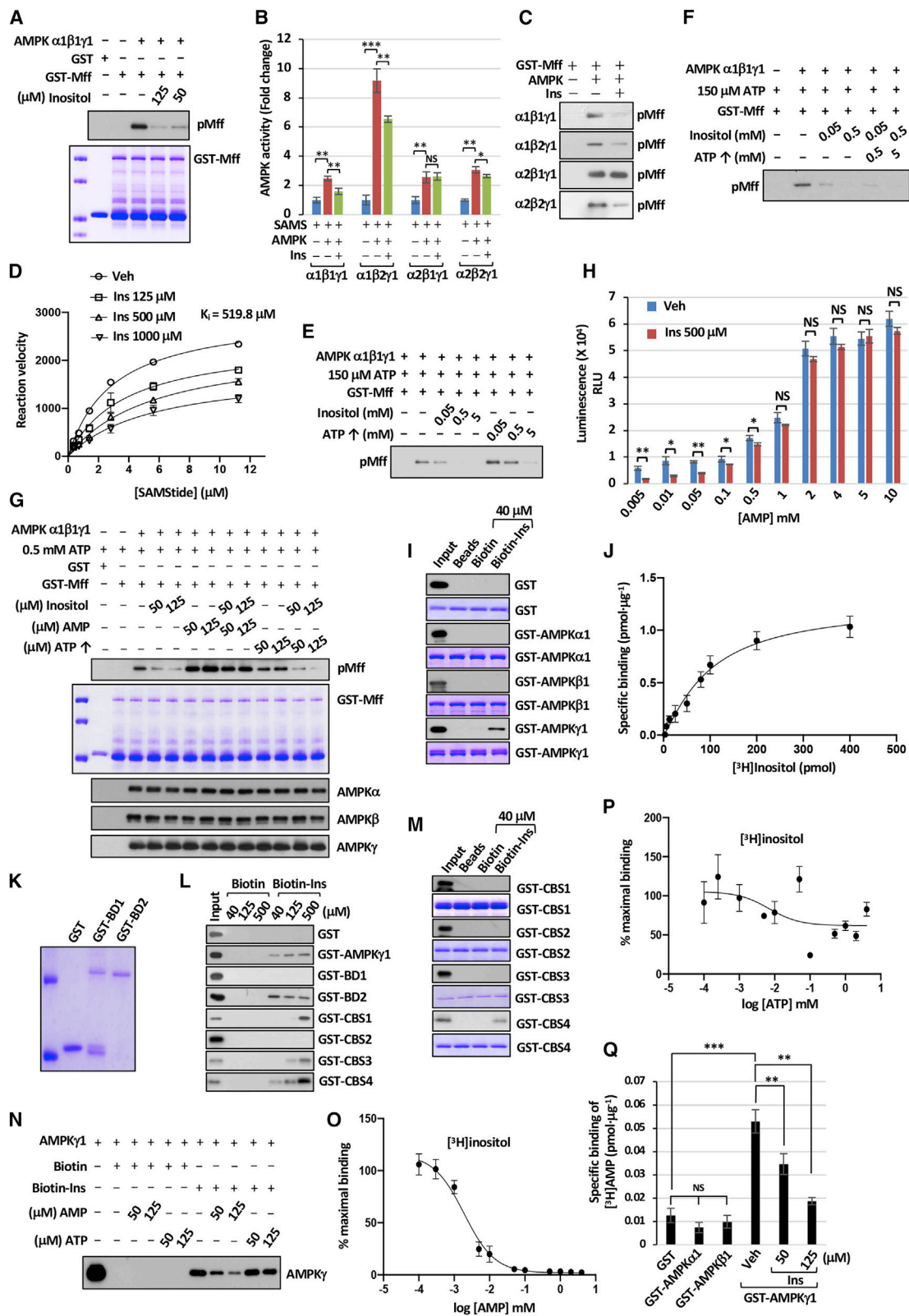
suggesting IMPA/inositol modulates diverse biological functions controlled by AMPK. Our study unveils that inositol is a previously uncharacterized metabolite directly restricting AMPK-dependent mitochondrial fission.

AMPK is activated in cells under energy stress conditions and inhibits mitochondrial ETC through an increased AMP level (Hardie, 2011). Our study reveals that in addition to AMP elevation, inositol decline is also a prerequisite step for AMPK activation upon the stimulation of these stress conditions. Inositol reduction results in unoccupied AMPK γ CBS1, CBS3, and CBS4 motifs, thereby allowing for AMP-AMPK γ binding to facilitate AMPK activation and mitochondrial fission (Figure S11). Hence, the AMP/inositol ratio is a critical determinant for AMPK activation. Notably, we identify IMPA1 and IMPA2 as key enzymes that maintain inositol biosynthesis, whose deficiency or inactivation is sufficient for AMPK activation and mitochondrial fission without affecting ATP level. It is important to note that the loss of CDIPT, a key enzyme depleting the inositol level in cells by converting inositol to PI, causes inositol accumulation that leads to the prevention of AMPK-dependent mitochondrial fission under energy stress conditions. Thus, inositol is a key rate-limiting metabolite that serves as an endogenous suppressor to restrict AMPK activation and mitochondrial fission in normal steady-state conditions.

The AMPK γ subunit contains four tandem repeats of a sequence motif termed the CBS repeat. There are three AMP binding sites (site-1, site-2, and site-3) where AMP binds to aspartic acid located on CBS1, CBS3, and CBS4 motifs, respectively (Xiao et al., 2007). Two AMP-1 and AMP-2 sites in which AMP binds to CBS1 and CBS3 motifs can be outcompeted by ATP and are therefore involved in AMP/ATP sensing. However, because there is no competition for ATP in the AMP binding site-3 in which AMP tightly binds to the CBS4 domain, ATP is not involved in AMP/ATP sensing (Xiao et al., 2007). Interestingly, AMP binding site-3 is tightly bound between the CBS4 motif and AMP due to a serine residue (315) interacting with a phosphate group that does not occur in the other two sites (Xiao et al., 2007). Our results further uncover that inositol binds to the CBS4 motif of AMPK γ with a much higher affinity than CBS1 and CBS3 motifs of AMPK γ . As inositol binds to the CBS4 motif of AMPK γ and competes with AMP for AMPK γ binding, it might also compete at the AMP binding site-3. Moreover, because 50 and 125 μ M of inositol is sufficient to repress AMPK activity and inhibition of AMPK activity by ATP requires up to 5 mM, it is plausible that AMPK may sense the inositol gradient more efficiently and rapidly than ATP in response to diverse energy stresses. As both inositol and ATP could repress AMPK activity under their physiological concentrations, AMPK could sense both inositol

Figure 6. CDIPT loss leading to inositol accumulation prevents mitochondrial fission upon energy stress

- (A) The levels of inositol in shLuc. or shCDIPT DU145 with or without glucose for 30 min were determined by using a K-INOSL assay kit. Three independent experiments were performed for the statistic of two-tailed unpaired *t*-test. **, *P* < 0.001.
- (B) Immunoblotting of shLuc. or shCDIPT DU145 cells with or without GD for 30 min.
- (C) Immunoblotting of shLuc. or shCDIPT PC3 cells with vehicle (Veh), GD, or Anti A treatment.
- (D) Confocal images of mitochondrial morphology in shLuc. or shCDIPT DU145 cells upon GD for 30 min. Scale bars, 20 μ m.
- (E) Quantification of the mitochondrial morphology of the cells shown in (D). **, *P* < 0.001; NS, non-significant by two-tailed unpaired *t*-test.
- (F) Confocal images of mitochondrial morphology in shLuc. or shCDIPT DU145 cells upon GD or lithium chloride treatment for 30 min.
- (G) Quantification of the mitochondrial morphology of the cells shown in (F). **, *P* < 0.001; ***, *P* < 0.0001; NS, non-significant by two-tailed unpaired *t*-test.



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and the ATP gradient for its activation in response to diverse energy stresses and mitochondrial damage (Figure S11).

In summary, our study provides compelling evidence that inositol is a second messenger that directly binds to AMPK γ to limit AMPK activation for mitochondrial fission induction. This study expands our current understanding of how inositol regulates novel signaling and biological processes beyond its classic role in PI cycle regulation. We identify AMPK as an inositol sensor detecting the inositol gradient in cells to adjust its activity for regulating mitochondrial dynamics. As deregulation of mitochondrial dynamics underlies various human diseases, we speculate that certain human diseases associated with IMPA1/2 or CIDIPT mutations (Figueiredo et al., 2016; Murphy et al., 2011; Ohnishi et al., 2014; Thakur et al., 2011) may result from aberrant AMPK-dependent mitochondrial fission caused by the alteration of the inositol level. We propose that inositol treatment, activating IMPA1/2, and/or targeting CIDIPT may be promising strategies for mitigating various human diseases associated with abnormal AMPK-dependent mitochondrial dynamics.

LIMITATION OF STUDY

Although our *in vitro* results suggest that inositol allosterically inhibits AMPK enzymatic activity by directly competing with AMP for binding to the AMPK γ subunit, it is not crystal clear how this competition occurs. To gain deeper insight into how inositol competes with AMP for binding to AMPK γ , identification of the crystal structure of the inositol-AMPK γ complex will be necessary. Another limitation concerning the role of IMPA1/inositol in restricting AMPK-mediated mitochondrial fission in our study is that we mostly conducted these investigations in cell-based models. However, the role of inositol in suppressing AMPK-mediated mitochondrial fission was not fully verified in *in vivo* animal experiments. Further *in vivo* experiments using conditional *Impa1* knockout mice to examine elevated AMPK activation and mitochondrial fission in diverse tissues will be required to further strengthen the conclusion.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.molcel.2021.08.025>.

Figure 7. Inositol competes with AMP for binding to AMPK γ and represses AMPK activity

- (A) AMPK activity upon Ins treatment was determined by immunoblotting with pMff.
- (B) AMPK activity was determined by ADP-Glo kinase assay.
- (C) AMPK activity was determined by immunoblotting with pMff.
- (D) The effects of Ins on AMPK activity by using SAMS peptide were determined by the fitting Michaelis-Menten equation. K_i for inositol was calculated by using the Michaelis-Menten equation.
- (E) AMPK activity with a fixed amount of ATP upon Ins or ATP treatment was determined by immunoblotting with pMff.
- (F) AMPK activity with a fixed amount of ATP upon adding Ins with or without ATP was determined by immunoblotting with pMff.
- (G) AMPK activity upon adding Ins, AMP, or ATP was determined by immunoblotting with pMff.
- (H) Bar graphs represent each concentration of AMP with or without Ins in AMPK enzymatic kinetic curve.
- (I and M) Immunoblotting of biotin pull-down assay incubated with recombinant fusion proteins.
- (J) Saturation binding of [3 H]inositol with recombinant GST-AMPK γ 1 proteins. A dissociation constant (K_D) value of $98.2 \text{ pmol} \pm 2.8 \text{ pmol}$ was determined for binding affinity between Ins and AMPK γ 1.
- (K) Coomassie blue staining of the indicated recombinant proteins GST-Bateman domain 1 (BD1) and Bateman domain 2 (BD2).
- (L) The binding between Biotin-Ins and indicated recombinant proteins was determined by immunoblotting.
- (N) The competitive binding assay was subjected to immunoblotting with a specific AMPK γ antibody.
- (O) Competition binding of [3 H]inositol ($3.75 \text{ } \mu\text{M}$) and AMP with recombinant GST-AMPK γ 1 protein (100 ng). $\text{IC}_{50} = 1.9 \text{ } \mu\text{M} \pm 0.5 \text{ } \mu\text{M}$.
- (P) Competition binding of [3 H]inositol ($3.75 \text{ } \mu\text{M}$) and ATP with recombinant GST-AMPK γ 1 protein.
- (Q) The competitive binding assay between AMP and Ins was determined upon incubation of $3.33 \text{ } \mu\text{M}$ of tritium-labeled AMP (^3H -AMP) and indicated recombinant proteins. *, $P < 0.05$; **, $P < 0.001$; ***, $P < 0.0001$; NS, non-significant by two-tailed unpaired *t*-test.

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AUTHOR CONTRIBUTIONS

C.-C.H. and H.-K.L. designed experiments and wrote the manuscript. C.-C.H. performed experiments and analyzed the data for all figures. H.G. and Y.J. analyzed MS-based metabolic profiling. X.Z., G.W., W.Z., Z.C., B.-S.P., C.X., G.J., R.K.M., W.Y., J.S., T.C., E.L., A.K., R.E., Z.-G.X., Z.C., and H.-Y.L. provided technical and editing support, comments, and suggestions.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
IMPA1 (clone H-7)	Santa Cruz Biotechnology	Cat#sc-374234; RRID: AB_10988246
Mff (clone B-2)	Santa Cruz Biotechnology	Cat# sc-398617; RRID: AB_2744543
phospho-Mff Ser146 (polyclone)	Cell Signaling Tech	Cat#49281; RRID: AB_2799354
IMPA2 (polyclone)	GeneTex	Cat#GTX103636; RRID: AB_1950587
AMPK α (polyclone)	Cell Signaling Tech	Cat#2532; RRID: AB_330331
AMPK β (polyclone)	Cell Signaling Tech	Cat#12063; RRID: AB_2797812
AMPK γ (polyclone)	Cell Signaling Tech	Cat#4187; RRID: AB_10695248
phospho-AMPK α T172 (clone D4D6D)	Cell Signaling Tech	Cat#50081; RRID: AB_2799368
PKA (clone A-2)	Santa Cruz, sc-28315	Cat#sc-28315; RRID: AB_628136
phospho-Drp1 S616 (polyclone)	Cell Signaling Tech	Cat#3455; RRID: AB_2085352
phospho-Drp1 S635 (clone D3A4)	Cell Signaling Tech	Cat#6319; RRID: AB_10971640
Fis (clone B-5)	Santa Cruz Biotechnology	Cat#sc-376447; RRID: AB_11149382
Mfn1 (clone D-10)	Santa Cruz Biotechnology	Cat#sc-166644; RRID: AB_2142616
Mfn2 (clone F-5)	Santa Cruz Biotechnology	Cat#sc-515647; RRID: AB_2811176
ULK (clone D8H5)	Cell Signaling Tech	Cat#8054; RRID: AB_11178668
Raptor (clone 24C12)	Cell Signaling Tech	Cat#2280; RRID: AB_561245
phospho-ULK S555 (clone D1H4)	Cell Signaling Tech	Cat#5869; RRID: AB_10707365
phospho-Raptor S792 (polyclone)	Cell Signaling Tech	Cat#2083; RRID: AB_2249475
Actin (clone AC-15)	Sigma-Aldrich	Cat#A5441; RRID: AB_476744
GAPDH (clone GAPDH-71.1)	Sigma-Aldrich	Cat#G8795; RRID: AB_1078991
Tubulin (clone AA13)	Sigma-Aldrich	Cat#T8203; RRID: AB_1841230
CDIPT (clone C-2)	Santa Cruz Biotechnology	Cat#sc-514255; RRID: AB_2893146
GST (clone B-14)	Santa Cruz Biotechnology	Cat#sc-138; RRID: AB_627677
LC3 α/β (polyclone)	Novus	Cat#NB100-2220; RRID: AB_10003146
Calcineurin (D-1)	Santa Cruz Biotechnology	Cat#sc-373803; RRID: AB_10917925
MiD51 (clone A-6)	Santa Cruz Biotechnology	Cat#sc-514135; RRID: AB_2893143
CREB (clone 24H2B)	Santa Cruz Biotechnology	Cat#sc-271; RRID: AB_627300
phospho-CREB S133 (clone 10E9)	Santa Cruz Biotechnology	Cat#sc-81486; RRID: AB_1125727
choactase (clone E-7)	Santa Cruz Biotechnology	Cat#sc-55557; RRID: AB_2291743
TOM20 (monoclonal)	Abcam	Cat#ab56783; RRID: AB_945896
HRP-coupled antibodies to mouse	Santa Cruz Biotechnology	Cat#sc-2055; RRID: AB_631738
HRP-coupled antibodies to rabbit	Thermo Fisher	Cat#31480; RRID: AB_228457
Alexa Fluor 488 anti-mouse IgG	Thermo Fisher	Cat#A-21202; RRID: AB_141607
Rabbit normal IgG	Santa Cruz Biotechnology	Cat#sc-2027; RRID: AB_737197
Mouse Anti-rabbit IgG	Cell Signaling Tech	Cat#45262; RRID: AB_2799281
Chemicals and recombinant proteins		
lithium chloride	Millipore	Cat#5910
AICAR	Sigma-Aldrich	Cat#A9978
A769662	Reagents Direct	Cat#A769662
Compound C	Sigma-Aldrich	Cat#P5499
Myo-inositol	Sigma-Aldrich	Cat#I7508
AMP	Sigma-Aldrich	Cat#A1752
ATP	Sigma-Aldrich	Cat#A2383

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
BIO	Selleckchem.com	Cat#S7198
L- α -Phosphatidylinositol	Sigma-Aldrich	Cat#P6636
myo-Inositol 1,4,5-trisphosphate hexapotassium salt	Tocris	Cat#1482
phosphatidylinositol 3,4,5-triphosphate	Echelon®	Cat#P-3924
AMPK α 1, β 1, γ 1 protein	Millipore	Cat#14-840
AMPK α 1, β 1, γ 1 protein	Promega	Cat#V1921
AMPK α 1, β 2, γ 1 protein	Thermo Fisher	Cat#PV6244
AMPK α 2, β 1, γ 1 protein	BPS Bioscience	Cat#40024
AMPK α 2, β 2, γ 1 protein	Thermo Fisher	Cat#PV6247
AMPK α 1	Novus Biologicals	Cat#NBP2-51992
AMPK β 1	Novus Biologicals	Cat#NBP2-23375
AMPK γ 1	Novus Biologicals	Cat#NBP-23377
ULK protein	Thermo Fisher	Cat# PV6430
SAMS peptide	Promega	Cat#V1921
Tetramethylrhodamine	Thermo Fisher	Cat# T669
Dulbecco's Modified Eagle's Medium w/o inositol	MP Biomedicals	Cat# 091642954
[3 H]AMP	American Radiolabeled Chemicals Inc.	Cat#ART0843
[3 H]inositol	American Radiolabeled Chemicals Inc.	Cat#ART0261

Critical commercial assays

K-INOSL Assay Kit	Megazyme	Cat#820517
Glucose-6-Phosphate Colorimetric Assay Kit	BioVision	Cat#K657
ATP Determination Kit	Thermal Fisher	Cat#A22066
ADP-Glo™ kinase assay	Promega	Cat#V6930

Experimental models: Cell lines

Human: PC3	ATCC	Cat#CRL-1435
Human: DU145	ATCC	Cat#HTB-81
Mouse: mouse embryonic fibroblasts	ATCC	Cat#SCRC-1008
Mouse: AMPK $^{-/-}$ mouse embryonic fibroblasts	Dr. Kun-Liang Guan (The University of California, San Diego)	N/A

Experimental models: Organisms/Strains

Mouse: B6SJL.SOD1-G93A	The Jackson Laboratory	JAX: 002726
Mouse: C57BL/6J	The Jackson Laboratory	JAX: 000664

Oligonucleotide

ShRNA: human IMPA1-#1	Sigma-Aldrich	Cat#TRCN0000310192
ShRNA: human IMPA1-#2	Sigma-Aldrich	Cat#TRCN0000296135
ShRNA: mouse IMPA1-#1	Sigma-Aldrich	Cat#TRCN0000081409
ShRNA: mouse IMPA1-#2	Sigma-Aldrich	Cat#TRCN0000306604
ShRNA: human IMPA2-#1	Sigma-Aldrich	Cat#TRCN0000050318
ShRNA: human IMPA2-#2	Sigma-Aldrich	Cat#TRCN0000050320
ShRNA: human CDIPT-#1	Sigma-Aldrich	Cat#TRCN0000035987
ShRNA: human CDIPT-#2	Sigma-Aldrich	Cat#TRCN0000035988
ShRNA: human INPP1-#1	Sigma-Aldrich	Cat#TRCN0000051600
ShRNA: human INPP1-#2	Sigma-Aldrich	Cat#TRCN0000051601
ShRNA: human PI4K2 α -#1	Sigma-Aldrich	Cat#TRCN0000195093
ShRNA: human PI4K2 α -#2	Sigma-Aldrich	Cat#TRCN0000037605
ShRNA: human PLC β 1-#1	Sigma-Aldrich	Cat#TRCN0000226441
ShRNA: human PLC β 1-#2	Sigma-Aldrich	Cat#TRCN0000226442
ShRNA: human AMPK α 1	Sigma-Aldrich	Cat#TRCN0000199831

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
ShRNA: human AMPK α 2	Sigma-Aldrich	Cat#TRCN0000219690
Recombinant DNA		
pGEX6P-1-AMPK α	This paper	N/A
pGEX6P-1-AMPK β	This paper	N/A
pGEX6P-1-AMPK γ	This paper	N/A
pGEX6P-1-Bateman domain 1	This paper	N/A
pGEX6P-1-Bateman domain 2	This paper	N/A
pGEX6P-1-CBS1	This paper	N/A
pGEX6P-1-CBS2	This paper	N/A
pGEX6P-1-CBS3	This paper	N/A
pGEX6P-1-CBS4	This paper	N/A
pFLAG-CMV-2-IMPA1	This paper	N/A
pFLAG-CMV-2-IMPA1 D220A	This paper	N/A
pFLAG-CMV-2-IMPA1 W219A	This paper	N/A
Software and algorithms		
ImageJ	ImageJ software	https://login.go.libproxy.wakehealth.edu/login?url=https://imagej.nih.gov%2fij%2findex.html
Fiji	Fiji software	https://imagej.net/fiji
Graphpad Prism 8	Graphpad software	N/A
CRISPR Design tool	Zhang Lab	https://zlab.bio/guide-design-resources
Deposited data		
Unprocessed images	This paper	https://dx.doi.org/10.17632/nvf6fgc77g.1

RESOURCE AVAILABILITY**Lead contact**

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Hui-Kuan Lin (hulin@wakehealth.edu).

Materials availability

The plasmids, stable cell lines and antibodies generated in this study have not been deposited to any repositories yet. These materials will be available upon request.

Data and code availability

- The unprocessed data of immunoblotting have been deposited to Mendeley Data: <https://dx.doi.org/10.17632/nvf6fgc77g.1> <http://data.mendeley.com/datasets/nvf6fgc77g/draft?a=3c37aafc-0338-44f4-8619-183d8fc4e67f>. The software utilized in this study can be found in the [Key resources table](#).
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS**Animal Studies**

C57BL/6J mice (Male, 14 weeks) were conducted for fasting and inositol treatment by intraperitoneal injection (30 mg/kg) or oral administration (8 mg/g) via drinking water. All the related protocols were approved animal protocol by Institutional Animal Care and Use Committee (IACUC) at Wake Forest School of Medicine.

Cell Culture and Viral Packaging

Wild-type (WT) and AMPK knockout (AMPK^{-/-}) mouse embryonic fibroblasts (MEFs) were cultured in DMEM medium supplied with 10% FBS, 2 mM glutamine or inositol-free DMEM medium (MP Biomedicals) supplied with 10% FBS, 2 mM glutamine. DU145 and

PC3 cells were cultured in RPMI-1640 medium supplied with 10% FBS, 2mM glutamine. To generate shRNA knockdown cells, HEK293T cells were prepared and co-transfected with either luciferase (shLuc.) or target gene shRNA with packaging plasmid (pPAX) and envelope plasmid (pMD.2G) by using the calcium phosphate transfection method. Medium was changed 6 hours later and virus particles were harvested after another 48 hours to infect parental cells, then selected by puromycin.

METHOD DETAILS

Immunoblotting, Immunoprecipitation and Immunofluorescence

Cells were lysed in RIPA buffer supplemented with proteinase inhibitor cocktail and subjected to immunoblotting (IB) by indicated antibodies. Immunoblotting assays were repeated at least three times for each dataset. For immunoprecipitation assays, primary antibodies were incubated with magic protein agarose A/G beads for 30 minutes at room temperature (RT), followed by incubation with cell lysates rotated at room temperature for 3 hours. The beads were washed four times with lysis buffer and analyzed by IB. Vehicle (NT), 25 μ M of inositol (Ins), 4 μ M of PI, IP₃ or PIP₃ treatment. For immunofluorescence, the cells were stained with 200 nM MitoTracker Red for 15 minutes, fixed for 10 min with 4% paraformaldehyde in phosphate buffered saline (PBS) and then permeabilized with 0.4% Triton X-100. The cells were then incubated with anti-Drp1 antibody (cell signaling, #14647) for 2–3 hours at room temperature and washed three times with phosphate-buffered saline, followed by incubation with Alexa 488 fluorescein-conjugated anti-mouse IgG. Nuclei were visualized by ProlongGold (Life Technologies). The stained cells were observed by Olympus FV1200 Spectral Laser Scanning Confocal Microscope.

Quantification of Mitochondrial Morphology

For quantification of mitochondrial morphology of MitoTracker-stained cells, scoring was done blindly to different genotypes and treatments. “Fragmented” was defined as cells with spherical mitochondria majority; “Short” was defined as cells with a majority of mitochondria less than about 10 μ m; “Long” represents cells which the majority of mitochondria was more than about 10 μ m, and “Fused” was defined as cells with highly interconnected mitochondria of less than 4–5 free ends. For the statistic of colocalization of Drp1 and MitoTracker, images were acquired in ImageJ using colocalization analysis. Colocalization were counted using particle analyzed and normalized to total mitochondrial area in each picture. Three independent experiments for each sample were included in statistics consisting of at least five pictures representing at least 100 cells in every condition. For quantification of mitochondrial morphology of transmission electron microscope, scoring was done blindly to different genotypes and treatments. “Short” was defined as the length of mitochondria less than about 500 nm; “Long” represents the length of mitochondria more than about 1 μ m. At least 50 mitochondria from the liver and skeletal muscle in each male mouse and three male mice for each group were counted and analyzed by an observer who is blind to the experimental conditions. To quantify the aspect ratio, elongation and interconnectivity of individual mitochondria, confocal images stained by MitoTracker Red FM using Fiji/ImageJ software were processed, binarized and subjected to particle analysis to generate mitochondrial morphology characteristics for each cell. Aspect ratio was employed as an index of major and minor axis of ellipse equivalent to the mitochondria. Mitochondrial elongation was measured by inverse circularity and mitochondrial interconnectivity was measured by the mean area to perimeter ratio according to the published protocol (Dagda et al., 2009; Wiemerslage and Lee, 2016). The mean area/mean perimeter ratio was employed as an index of mitochondrial interconnectivity. Inverse circularity was validated using the equation of $1/44 \times \pi \times (\text{mean area} / \text{mean perimeter}^2)$ as a measure of mitochondrial elongation.

Biotin Pull-Down Assay, [³H]AMP Binding Assay and *In Vitro* Kinase Assay

The full length recombinant proteins of GST-AMPK α , β , γ or GST-CBS1, CBS2, CBS3, CBS4 of AMPK γ were expressed in *Escherichia coli* BL21 and purified. The purity of the recombinant proteins was resolved by SDS-PAGE, followed by Coomassie blue staining. For biotin-inositol pull-down assays, magnetic Dynabeads MyOne Streptavidin T1 (Thermal Fisher) was pre-incubated with free biotin or biotin-labeled inositol for 30 minutes at room temperature (RT), and then incubated with cell lysates or recombinant proteins rotated for 2 hours at 4°C. The beads were washed with PBS 3–4 times and analyzed by Immunoblotting. For competition assay, 50 mM and 125 mM of ATP and AMP were incubated with AMPK γ recombinant protein and biotin-labeled inositol for 1 hour at 4°C, washed and analyzed by IB. For *in vitro* AMP and AMPK binding assay, [³H]AMP obtained from American Radiolabeled Chemicals Inc. was incubated with the GST fusion recombinant proteins on glutathione-Sepharose beads (GE Healthcare) or 50 mM and 125 mM of inositol, for competitive binding with AMP, at 4°C for 1 hour in binding buffer (10 mM HEPES pH 7.5, 50 mM NaCl, 0.1% Nonidet P-40, 0.5 mM dithiothreitol, and 0.5 mM EDTA), washed two times with PBS, and radioactivity was determined using a liquid scintillation counter (LS6500; Beckman Coulter). For *in vitro* kinase assay, the recombinant protein GST-Mff was expressed in *Escherichia coli* BL21 and purified. Recombinant protein ULK was obtained from Thermo Fisher, active AMPK from Millipore (#14-840). 100 ng of active AMPK incubated with 100 ng of recombinant proteins and 0.5 mM of ATP in kinase reaction buffer (50 mM HEPES pH 7.4, 1 mM MgCl₂, 0.1 mM dithiothreitol, and 0.1 mM EGTA) upon adding 50 and 125 μ M of inositol, 50 and 125 μ M of AMP or 50 and 125 μ M of ATP. The reaction was incubated at 30°C for 15 minutes and subjected to immunoblotting using phospho-Mff S146 or phospho-ULK S555 antibody. For *in vitro* SAMS peptide phosphorylation assay, the recombinant AMPK α 1 β 1 γ 1 (Promega, V1921), α 1 β 2 γ 1 (Thermo Fisher, PV6244), α 2 β 1 γ 1 (BPS Bioscience, 40024) or α 2 β 2 γ 1 (Thermo Fisher, PV6247) complex was obtained from Thermo fisher or BPS Bioscience. 100 ng of AMPK α 1 β 1 γ 1, α 1 β 2 γ 1, α 2 β 1 γ 1 or α 2 β 2 γ 1

(Thermo Fisher, PV6247) complex was incubated with fixed amount of ATP (150 μ M) and 400 ng of SAMS peptide (HMRSAMSGSLHLVKRR) (Promega, V1921) in kinase reaction buffer (8 mM Tris pH7.5, 4 mM MgCl₂, 20 μ g/ml BSA, 10 μ M dithiothreitol) upon adding 125 μ M of inositol at room temperature for 30 minutes. Flag vector control, Flag-AMPK γ 2 and Flag-AMPK γ 3 complexes were overexpressed in 293T cells followed by immunoprecipitation with Flag antibody and elution by Flag peptide. Active 100 ng of heterotrimer AMPK α 1 (Novus Biologicals, NBP2-51992), AMPK β 1 (Novus Biologicals, NBP2-23375) and Flag-AMPK γ 2 (α 1 β 1 γ 2) /Flag-AMPK γ 3 (α 1 β 1 γ 3) complex, Flag-AMPK γ 2 (AMPK γ 2) or Flag-AMPK γ 3 (AMPK γ 3) complexes was incubated with fixed amount of ATP (150 μ M) and 100 ng of GST-Mff or 400 ng of SAMS peptide, followed by adding 125 μ M of inositol at 30°C for 15 or 30 minutes. The AMPK kinase activity was determined by ADP-Glo™ kinase assay (Promega, V6930) according to manufacturer's procedures. The values of AMPK enzymatic kinetic curve were calculated based on the equation: $Y = \text{basal} + \{[(\text{activation} \times \text{basal} - \text{basal}) \times X] / (EC_{50} + X)\}$ and then normalized with basal, where Y is kinase activity and X is AMP, ATP or inositol concentrations. All data are representative of three independent experiments and means $\pm$ SEM of triplicate values.

AMPK Enzyme Kinetics

The velocity of AMPK enzymatic kinetics was calculated using the Michaelis-Menten Equation using a substrate concentration range of between 200 nM and 11 μ M with vehicle, 125, 500, 1000 μ M of inositol treatment. Lineweaver-Burk plots were generated from 1/velocity and 1/concentrations of SAMS peptide. The inhibition constant (K_i) of inositol was determined from Michaelis-Menten saturation plot versus the concentrations of SAMS peptide. Graphs were plotted by using Prism 8 (GraphPad Software Inc.) The values of the correlation coefficient, slope and standard errors were obtained by linear and non-linear regression analysis using these programs. Each condition is shown from three independent experiments (mean $\pm$ SEM, n = 3).

Saturation and Competition Binding Assay Using Tritium-labeled inositol and Isothermal Titration Calorimetry (ITC)

Recombinant GST-AMPK γ proteins and [³H]inositol were incubated at room temperature (RT) for 1 hour in binding buffer (10 mM HEPES pH 7.5, 50 mM NaCl, 0.1% Nonidet P-40, 0.5 mM dithiothreitol, and 0.5 mM EDTA), followed by glutathione-Sepharose beads (GE Healthcare) pull-down for 3 hours. 7-9 different concentrations of [³H]inositol in a range of 50 μ M to 4 mM were employed. After glutathione-Sepharose beads were washed three times with PBS, radioactivity was determined using a liquid scintillation counter (LS6500; Beckman Coulter). For competition assay, recombinant GST-AMPK γ proteins and [³H]inositol were incubated at room temperature (RT) for 1 hour in binding buffer upon adding AMP or ATP. 9-11 different concentrations of AMP or ATP in a range of 0.1 μ M to 4 mM were employed. After glutathione-Sepharose beads were washed three times with PBS, radioactivity was determined using a liquid scintillation counter. All data are representative of three independent experiments and means $\pm$ SEM of triplicate values. [³H] inositol obtained from American Radiolabeled Chemicals Inc. Equilibrium dissociation constant (K_d) was determined from saturation binding plot versus the concentrations of [³H]inositol. Graphs were plotted by using Prism 8 (GraphPad Software Inc.) The values of the correlation coefficient, slope and standard errors were obtained by linear and non-linear regression analysis using these programs. ITC experiments were performed using a MicroCal PEAQ-ITC calorimeter at 25°C. The sample cell was filled with 300 μ L of recombinant protein GST or GST-AMPK γ (7.8 μ M in 100 mM Tris buffer) and stirred at 750 rpm. The syringe was filled with 90 μ L inositol (780 μ M in 100 mM Tris buffer). Inositol solution was injected into the sample cell with a single injection of 0.4 μ L, followed by nineteen 1.5 μ L injections at 150 s intervals. The data were analyzed with the MicroCal PEAQ-ITC Analysis Software using the single binding site model.

Time-lapse Microscopy

DU145 cells were seeded in the chambered coverslips (LabTeK®II) and stained with 200 nM MitoTracker Red for 15 minutes before treatment of 25 μ M inositol (Sigma), 300 μ M A769662 (Selleckchem) or 2 mM AICAR. Image was collected and acquired every minute for 30 minutes by Olympus FV1200 Spectral Laser Scanning Confocal Microscope at 37°C and 5% CO₂. Each image was acquired, exported and uncompressed in AVI format with 2 times speed.

Lentiviral Infection and Sequences of shRNAs

To generate lentivirus-producing cell lines, calcium phosphate was used to transfect with luciferase (control), IMPA1, IMPA2, CDIPT, INPP1, PI4K2 α and PLC β 1 shRNA, virus Gag-Pol packing vector (pPAX) and virus envelope glycoprotein expression vector (pMD.2G) in 293T cells. Supernatants were collected 48 hours after transfection, filtered (0.45 μ m), purified by centrifugation, and stored at -80°C. Infections were carried out overnight in the presence of 8 μ g/mL Polybrene (Sigma). After viral infection, cells were selected with puromycin (2 μ g/ml). The individual mission shRNAs kept in frozen bacterial glycerol were purchased from Sigma Aldrich. The following shRNAs sequences were used:

IMPA1 for human #1 TRCN0000310192: CCGGCAACGAGACGACGAAGATTAACCTCGAGTTAATCTTCGTCGCTCGTTGTTTTTG;
 IMPA1 for human #2 TRCN0000296135: CCGGCATCAACTGAAATAGTCAATTCTCGAGAATTGACTATTTTCAGTTGATGTTTTTG;
 IMPA1 for mouse #1 TRCN0000081409: CCGGCGGAGAATAATTGCCGCAAATCTCGAGATTTGCCGCAATTATTCTCCGTTTTTG; IMPA1 for mouse #2 TRCN0000306604: CCGGATGTTCCAGAAGCCATATTTGCTCGAGCAAATATGGCTTCTGGAACATTTTTTG; IMPA2 #1 TRCN0000050318: CCGGCCTGAAGCTGTTCTGAGTAACCTCGAGTTACTCAGGAACAGCTTCAGGTTTTTG; IMPA2 #2 TRCN0000050320:

CCGGGCTGTTTCGACAAGAGCTTGAAGCTCGAGTTCAAGCTCTTGTGCGAACAGCTTTTTG; CDIPT #1 TRCN0000035987:
CCGGGAGTCACAAGATGATCGACTTCTCGAGAAGTCGATCATCTTGTGACTCTTTTTG; CDIPT #2 TRCN0000035988:
CCGGCTCGCGCTCTTAATCAAGGAAGCTCGAGTTCTTGATTAAGAGCGCGAGTTTTG; INPP1 #1 TRCN0000051600:
CCGGGCTTAGAAAGAAATCCAGAACTCGAGTTTCTGGATTTCTTTCTAAGCTTTTTG; INPP1 #2 TRCN0000051601:
CCGGCCACATTCAAATGGGACTTCTCGAGAAGAGTCCCATTTGAATGTGGTTTTG; PI4K2 α #1 TRCN0000195093:
CCGGCAATGACAAGTGGCTGATTAAGTTCGAGTTAATCAGCCAGTTGTCATTGTTTTG; PI4K2 α #2 TRCN0000037605:
CCGGCCATAAGCAGATTGCTGTCATCTCGAGATGACAGCAATCTGCTTATGGTTTTG; PLC β 1 #1 TRCN0000226441:
CCGGCAGCGAGATCCTCGGCTTAATCTCGAGATTAAGCCGAGGATCTCGCTGTTTTG; PLC β 1 #2 TRCN0000226442: CCGGT
CTAATCTGGTGAAGTATATTCTCGAGAATATAGTTCACCAGATTAGTTTTG; AMPK α 1 #1 TRCN0000199831:
CCGGGTGACCTCACTTGACTTCTCTCGAGAGAAGAGTCAAGTGAGGTCACTTTTTG; AMPK α 1 #2 TRCN0000219690:
CCGGGAAGGTTGTAACCCATATTACTCGAGTAATATGGGTTTACAACCTTCTTTTTG

Knockout Cell Generation Using CRISPR

Knockout cells were generated using the Cas9 nickase strategy as described previously (Ran et al., 2013). Two pairs of guide RNAs (gRNA) targeting the exon 3 and exon 4 respectively were designed for IMPA1 gene using the online design tool at <https://zlab.bio/guide-design-resources>. Each gRNA duplex was cloned into pX462 vector encoding SpCas9n-2A-puro (Addgene # 48141). The lentivirus of two pairs of guide RNAs was produced and used to infect cells, which were selected by puromycin (2 μ g/ml) and the single colony was picked. Individual clones with IMPA1 knockout were verified by immunoblotting.

Non targeting scramble gRNA sequence:

guide A: CGAGGCTTAACGCCAGATTC; guide B: GAACTCGTTAGGCCGTGAAG

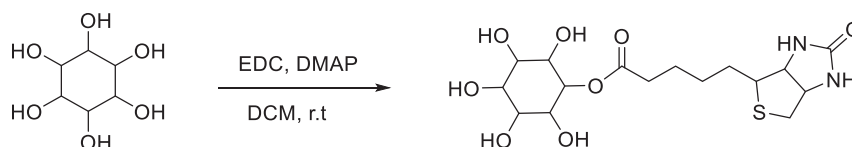
IMPA1 gRNA sequence:

guide A: AATAGGGTCAATGATCCATG; guide B: CCTTTTGTAGCTGTTTCAAT

Seahorse XFp Cell Mito Stress Test

Seahorse XF Cell Mito Stress Tests (Seahorse Bioscience, Agilent Technologies) were performed according to the manufacturer's protocol on an XF instrument. Briefly, one day prior to assay, the sensor cartridge was hydrated with Seahorse XF Calibrant buffer pH 7.4 (Seahorse Bioscience) at 37°C without CO₂ overnight. PC3 and DU145 (15,000 cells /well) were plated in Seahorse XF Cell Culture microplates (Seahorse Bioscience) in RPMI-1640 medium overnight. At day of assay, XF assay medium was supplemented with 10 mM glucose, 1 mM pyruvate, and 2 mM glutamine with pH adjusted to 7.4. Cells were rinsed twice with XF assay medium and incubated with the fresh XF assay medium for 1 hour in 37 °C without CO₂. After placing the calibration plate with loaded sensor of cartridge, 1 μ M of Oligomycin, 0.5 μ M of FCCP and 1 μ M of Rotenone/Antimycin A were added in the ports respectively, and oxygen consumption rate (OCR) was measured by Seahorse XF Analyzer. Data analysis was performed using the Seahorse XF Wave software.

The Synthesis of Biotin-inositol



A solution of inositol (180 mg, 1.0 mmol), 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC) (384 mg, 2.0 mmol) and 4-dimethylaminopyridine (DMAP) (11 mg, 0.01 mmol) in dry dichloromethane (DCM) (10 mL) was stirred at room temperature. Then, biotin (244 mg, 1.0 mmol) in 10 mL DCM was added dropwise for 5 min. The mixture was stirred at room temperature overnight. The solvent was removed and the product was precipitated from 20% ethyl acetate/hexanes. 210 mg of wax solid was obtained and the yield was 52%. LC/MS calculated for C₁₆H₂₇N₂O₈S [M+H]⁺, 407; found 407.

Inositol level, ATP level, G6P level, Mitochondrial ROS and Membrane Potential Measurement

5 × 10⁶-10⁷ cells were harvested and extracted by 80% methanol. The inositol, glucose-6-phosphate (G6P) levels and ATP levels in cells were determined by the K-INOSL Assay Kit (Megazyme, 820517), Glucose-6-Phosphate Colorimetric Assay Kit (BioVision, K657) and ATP Determination Kit (Thermal Fisher, A22066), respectively, according to manufacturer's protocol. The inositol, G6P and ATP levels were normalized by protein concentration in each experimental group. The inositol, G6P and ATP levels are shown as mean ± SEM of three independent experiments. Mitochondrial ROS levels were measured using the MitoSOX probe

(Life Technologies) according to the manufacturer's protocol. Following treatment with 2.5 μ M of MitoSOX for 30 minutes, cells were trypsinized, resuspended in PBS and immediately analyzed by flow cytometry. Mitochondrial membrane potential levels were measured using Tetramethylrhodamine (TMRM) (ThermoFisher) according to the manufacturer's protocol. Following treatment with 200 nM of TMRM for 30 minutes at 37°C, cells were trypsinized, resuspended in PBS and immediately analyzed by flow cytometry.

Extraction of Metabolites and LC-MS/MS Measurement

For the extraction of Ins, AMP, ADP, ATP, IP, 1×10^6 cells were harvested from the 10 cm dish. After culture media were aspirated, cells were washed twice in ice-cold PBS and lysed in 1 mL of ice-cold 80% methanol with the plates placed on dry ice to quench metabolism. After 5 minutes, cells were scraped and transferred into 1.5 mL tubes and incubated in -80°C for 30 minutes. Lysates were centrifuged at 13,000 rpm for 10 minutes and supernatants were transferred to new vials and dried at 30°C in a Speed-Vac for 3 hours. For the extraction of PI, PIP, PIP₂, PIP₃, IP₂, IP₃, 1×10^6 cells were harvested from the 10 cm dish. After culture media were aspirated, cells were washed twice in ice-cold PBS and lysed in 1 mL of ice-cold 25% methanol containing 20 μ M PC (17:0) and PG (17:0) with the plates placed on dry ice to quench metabolism. Cells were scraped and transferred into 1.5 mL tubes and kept in MTBE (MTBE:methanol:H₂O = 10:2:5). After sonicating in ice bath for 15 minutes, cells were incubated in -20°C for 30 minutes, followed by centrifugation at 3,000 rpm for 5 minutes. Supernatants evaporated the MTBE layer and were dried in 100 μ L of chloroform:MeOH (1:1). The targeted LC-MS/MS method was modeled and developed as previously described (Carroll et al., 2015; Eghlimi et al., 2020). Briefly, all LC-MS/MS experiments were performed on an Agilent 1290 UPLC-6490 QQQ-MS (Santa Clara, CA) system. Each sample was injected twice, 10 μ L for analysis using negative ionization mode and 4 μ L for analysis using positive ionization mode. Both chromatographic separations were performed in hydrophilic interaction chromatography (HILIC) mode on a Waters XBridge BEH Amide column (150 $\times$ 2.1 mm, 2.5 μ m particle size, Waters Corporation, Milford, MA). The flow rate was 0.3 mL/min, auto-sampler temperature was kept at 4 C, and the column compartment was set at 40 C. The mobile phase was composed of Solvents A (10 mM ammonium acetate, 10 mM ammonium hydroxide in 95% H₂O/5% ACN) and B (10 mM ammonium acetate, 10 mM ammonium hydroxide in 95% ACN/5% H₂O). After the initial 1 min isocratic elution of 90% B, the percentage of Solvent B decreased to 40% at $t = 11$ min. The composition of Solvent B maintained at 40% for 4 min ($t = 15$ min), and then the percentage of B gradually went back to 90%, to prepare for the next injection. The mass spectrometer is equipped with an electrospray ionization (ESI) source. Targeted data acquisition was performed in multiple-reaction-monitoring (MRM) mode. The whole LC-MS system was controlled by Agilent MassHunter Workstation software (Santa Clara, CA). The extracted MRM peaks were integrated using Agilent MassHunter Quantitative Data Analysis (Santa Clara, CA).

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical significance was identified by Student's *t* test using Prism 8 software. *P* values of less than 0.05 were considered statistically significant; *, $p < 0.05$; **, $p < 0.001$; ***, $p < 0.0001$; NS, non-significant.